NEET 2015

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION



All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS 360

Physics

Q. 1 The position vector of a particle $\vec{\mathrm{R}}$ as a function of time is given by :

$\vec{\mathrm{R}} = 4\sin(2\pi t)\,\hat{i} + 4\cos(2\pi t)\,\hat{j}$, where R is in meters, t is in seconds, $\hat{i}$ and $\hat{j}$ denote unit vectors along *x-and y*-directions, respectively. Which one of the following statements is wrong for the motion of a particle?

Option 1:

Magnitude of acceleration vector is $\frac{v^2}{\mathrm{R}}$ where v is the velocity of particle.

Option 2:

Magnitude of the velocity of particle is 8 meter/ second

Option 3:

Path of the particle is a circle of radius 8 meter.

Option 4:

Acceleration vector is along $-\vec{\mathrm{R}}$.

Correct Answer:

Acceleration vector is along $-\vec{\mathrm{R}}$.

Solution:

Position $\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$

Velocity: $\vec{v} = 4\times 2\pi\cos(2\pi t)\hat{i} - 4\times 2\pi\sin(2\pi t)\hat{j}$

Acceleration: $a = -4(2\pi)^2\sin(2\pi t)\hat{i} - 4(2\pi)^2\cos(2\pi t)\hat{j} = -(2\pi)^2\vec{R}$

$\therefore$ acceleration is along $-\vec{R}$

Magnitude of velocity:

Magnitude of acceleration:

$$|\vec{a}| = \left|-(2\pi)^2\vec{R}\right| = 4\pi^2|\vec{R}| = 16\pi^2 = \frac{(8\pi)^2}{4} = \frac{v^2}{R}$$

$R_x = x = 4\sin(2\pi t)$

$R_y = y = 4\cos(2\pi t)$

$$R_x^2 + R_y^2 = x^2 + y^2 = 4^2$$

The path of the particle is a circle of radius 4.

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Hence, the answer is the option (4).

Q. 2 The energy of the em waves is of the order of 15 ke V. To which part of the spectrum does it belong ?

Option 1:
Infra - red rays

Option 2:
Ultraviolet rays

Option 3:
γ - rays

Option 4:
X - rays

Correct Answer:
X - rays

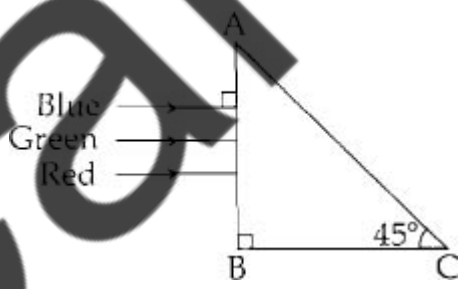
Solution:
The wavelength corresponding to 15KeV

$$\lambda = \frac{12400}{15KeV}\mathring{A} = \frac{12.40}{15}\mathring{A} = 0.83\mathring{A}$$

This belongs to the region of X-rays.

Hence, the answer is the option (3).

Q. 3 A beam of light consisting of red, green and blue colours is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47, respectively.



The prism will:

Option 1:
separate all the three colours from one another

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Option 2:
not separate the three colours at all

Option 3:
separate the red colour part from the green and blue colours

Option 4:
separate the blue colour part from the red and green colours

Correct Answer:
separate the red colour part from the green and blue colours

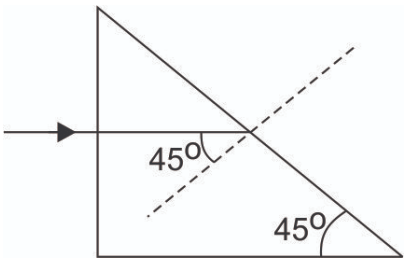
Solution:
As we learnt in

Critical angle -

$\sin c$ = Refractive index of rarer medium / Refractive index of denser medium

- wherein

When angle of incidence of a travelling from a dence medium to rarer medium is greater than critical angle, no refraction occurs.



angle of incidence at surface AC= 45°

Critical Angle of incidence

$$sin\ c = \frac{1}{\mu}$$

For red $\mu_r = 1.39$

For green $\mu_g = 1.44$

For blue $\mu_b = 1.47$

Since refractive index of green and blue is greater than $\sqrt{2}$ and hence their critical angles are less than 45°

So green and blue will be completely reflected while red will emerge out of prism. Hence red will be separated from green and blue.

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Q. 4 Two particles A and B, move with constant velocities $\vec{v_1}$, $\vec{v_2}$. At the initial moment, their position vectors are $\vec{r_1}$ and $\vec{r_2}$ respectively. The condition for particles A and B for their collision is :

Option 1:
$\vec{r_1} \cdot \vec{v_1} = \vec{r_2} \cdot \vec{v_2}$

Option 2:
$\vec{r_1} \times \vec{v_1} = \vec{r_2} \times \vec{v_2}$

Option 3:
$\vec{r_1} - \vec{r_2} = \vec{v_1} - \vec{v_2}$

Option 4:
$\frac{\vec{r_1} - \vec{r_2}}{\left|\vec{r_1} - \vec{r_2}\right|} = \frac{\vec{v_2} - \vec{v_1}}{\left|\vec{v_2} - \vec{v_1}\right|}$

Correct Answer:
$\frac{\vec{r_1} - \vec{r_2}}{\left|\vec{r_1} - \vec{r_2}\right|} = \frac{\vec{v_2} - \vec{v_1}}{\left|\vec{v_2} - \vec{v_1}\right|}$

Solution:
Let the particles A and B collide at time t. For their Collision, the position vectors of both particles should be the same at time t.

i.e. $\vec{r_1} + \vec{v_1}t = \vec{r_2} + \vec{v_2}t$

$$\vec{r_1} - \vec{r_2} = \vec{v_2}t - \vec{v_1}t$$
$$\Rightarrow \left|\vec{r_1} - \vec{r_2}\right| = \left|\vec{v_2} - \vec{v_1}\right| t$$
$$t = \frac{\left|\vec{r_1} - \vec{r_2}\right|}{\left|\vec{v_2} - \vec{v_1}\right|}$$

For the two bodies to collide their relative position vector and relative velocity should be along the same direction.

$$\Rightarrow \frac{\vec{r_1} - \vec{r_2}}{\left|\vec{r_1} - \vec{r_2}\right|} = \frac{\vec{v_2} - \vec{v_2}}{\left|\vec{v_2} - \vec{v_1}\right|}$$

Hence, the answer is the option (4).

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Q. 5 At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is :

Option 1:
$\frac{\pi}{2}$ radian

Option 2:
π radian

Option 3:
$\frac{\pi}{8}$ radian

Option 4:
$\frac{\pi}{4}$ radian

Correct Answer:
π radian

Solution:
Intensity of Diffracted light -

$$I = I_0 \frac{\sin^2 \beta}{\beta^2}$$

And

$$\beta = \frac{\pi}{\lambda} (b \sin \theta)$$

At the first minimum, wathe valet coming from the edge of the slit is balanced out by the wavelet coming from the middle of the slit. Hence the phase difference between the wavelet from the edge of the slit to the wavelet from the middle of the slit is π radian.

Hence, the answer is the option (2).

Q. 6 A proton and an alpha particle both enter a region of uniform magnetic field B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by the proton is 1 MeV, the energy acquired by the alpha particle will be:

Option 1:
0.5 MeV

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Option 2:
1.5 MeV

Option 3:
1 MeV

Option 4:
4 MeV

Correct Answer:
1 MeV

Solution:
The radius of charged particle -

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$$

$$R = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

$$K_p = \frac{q_p^2 B^2 R_p^2}{2m_p},\ K\alpha = \frac{q_\alpha^2 B^2 R_\alpha^2}{2m_\alpha}$$

$$K = \frac{q^2 B^2 R^2}{2m}$$

$$\therefore \frac{K\alpha}{K_p} = \left(\frac{q_\alpha}{q_p}\right)^2 \left(\frac{m_p}{m_\alpha}\right) \left(\frac{R_p}{R_\alpha}\right)^2$$

$$\Rightarrow K_\alpha = K_p \left(\frac{q_\alpha}{q_p}\right)^2 \left(\frac{m_p}{m_\alpha}\right) \left(\frac{R_\alpha}{R_p}\right)^2 = (2)^2(\frac{1}{4})(1)^2 = 1MeV$$

Hence, the answer is the option (3).

Q. 7 A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be:

Option 1:
0.25 A

Option 2:
2 A

Option 3:
1 A

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Option 4:
0.5 A

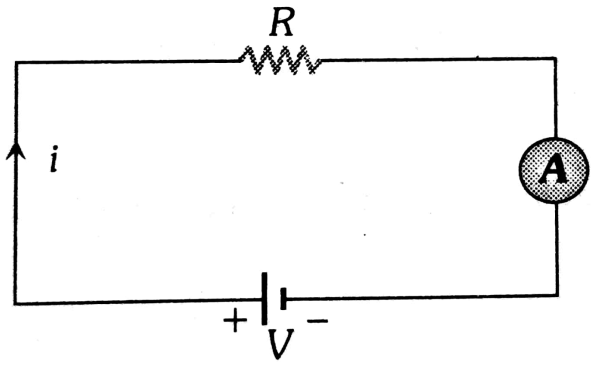
Correct Answer:
0.5 A

Solution:
As we learnt in

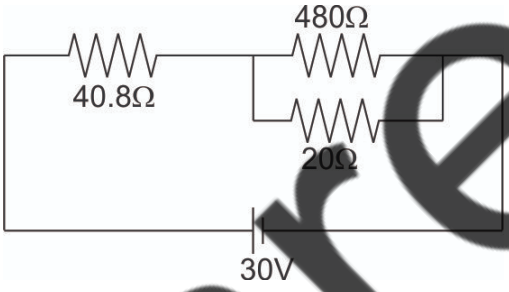
Ammeter -

It is a device used to measure current and always connected in series

- wherein



Resistance of ammeter is $R_A = \frac{(480\Omega)(20\Omega)}{(480\Omega)+(20\Omega)} = 19.2\Omega$



As ammeter is in series with 40.8 Ω

Total resistance of the circuit

R=40.8Ω$+R_A$ = 40.8Ω+19.2Ω

From omni's law

$$I = \frac{V}{R} = \frac{30V}{60\Omega} = \frac{1}{2}A = 0.5A$$

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Q. 8 The value of the coefficient of volume expansion of glycerin is 5 x 10^{-4} K^{-1}. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is:

Option 1:
0.020

Option 2:
0.025

Option 3:
0.010

Option 4:
0.015

Correct Answer:
0.020

Solution:

$\gamma = 5 \times 10^{-4} K^{-1}$

$\rho = \frac{\rho_o}{1+\gamma\Delta T} = \rho_o(1-\gamma\Delta T)$

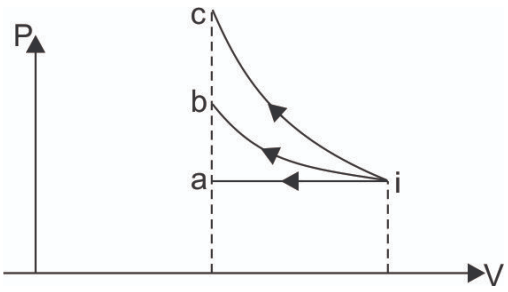
$\Rightarrow \rho = \rho_o - \rho_o\gamma\Delta T \quad or \quad \frac{\rho_o - \rho}{\rho_o} = \gamma\Delta T$

$\frac{\rho_o - \rho}{\rho_o} = Fractional\ change\ in\ the\ density\ of\ glycerin = \gamma\Delta T = 5 \times 10^{-4} \times 40 = 0.02$

Hence, the answer is the option (1).

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Q. 9 An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas ?



$ia \rightarrow Isobaric$

$ib \rightarrow Isothermal$

$ic \rightarrow Adiabatic$

Option 1:
Isobaric

Option 2:
Isochoric

Option 3:
Isothermal

Option 4:
Adiabatic

Correct Answer:
Adiabatic

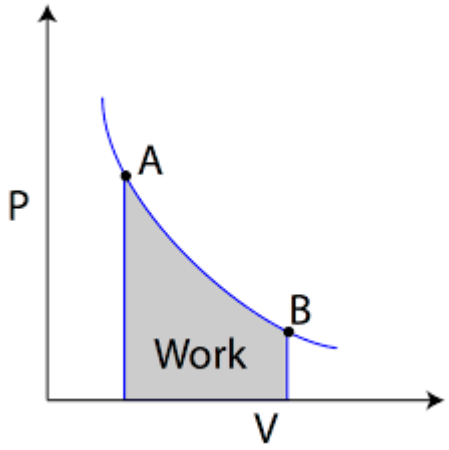
Solution:
As we learnt in

Work by area -

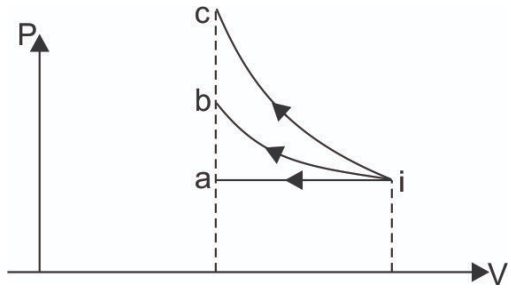
If we plot a process on PV diagram.Then area under the curve will give work done.

- wherein

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All the process are shown in the given figure.



$ia \rightarrow Isobaric$

$ib \rightarrow Isothermal$

$ic \rightarrow Adiabatic$

Work done is area under PV diagram

Maximum area covered under ic - Hence maximum work done during adiabetic process.

Correct option is 4.

Q. 10 A series R-C circuit is connected to an alternating voltage source. Consider two situations:

a. When capacitor is air filled.
b. When capacitor is mica filled.

Current through resistor is *i* and voltage across capacitor is V then :

Option 1:

$V_a > V_b$

Option 2:

$i_a > i_b$

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Option 3:

$V_a = V_b$

Option 4:

$V_a < V_b$

Correct Answer:

$V_a > V_b$

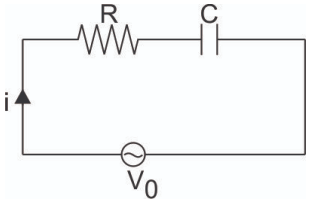
Solution:

As we learnt in

Impedence -

$$Z = \sqrt{X_C^2 + R^2} = \sqrt{R^2 + \left(\frac{1}{\omega c}\right)^2}$$

-



Current through resister i= current in the circuit

$$\frac{V_o}{\sqrt{R^2 + X_c^2}} = \frac{V_o}{\sqrt{R^2 + \left(\frac{1}{wc}\right)^2}}$$

Voltage across capacitor $V = iX_c$

$$= \frac{V_o}{\sqrt{R^2 + \left(\frac{1}{wc}\right)^2}} \times \frac{1}{wc} = \frac{V_o}{\sqrt{R^2 w^2 c^2 + 1}}$$

$As,\ \ C_a < C_b$

$\therefore i_a < i_b \ \ and \ \ V_a > V_b$

Q. 11 Light of wavelength 500 nm is incident on a metal with work function 2.28 eV. The de Broglie wavelength of the emitted electron is:

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Option 1:

$< 2.8 \times 10^{-9}\mathrm{m}$

Option 2:

$\geqslant 2.8 \times 10^{-9}\mathrm{m}$

Option 3:

$\leqslant 2.8 \times 10^{-12}\mathrm{m}$

Option 4:

$< 2.8 \times 10^{-10}\mathrm{m}$

Correct Answer:

$\geqslant 2.8 \times 10^{-9}\mathrm{m}$

Solution:

As we learnt in

Conservation of energy -

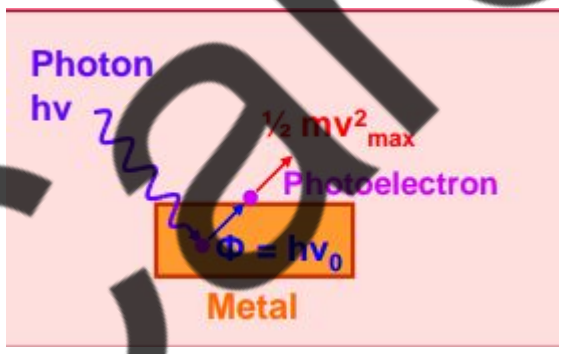
$$h\nu = \phi_0 + \frac{1}{2}mv_{max}^2$$

$$h\nu = h\nu_0 + \frac{1}{2}mv_{max}^2$$

$$h\left(\nu - \nu_0\right) = \frac{1}{2}mv_{max}^2$$

$where, h - Plank's constant\ \nu - Frequency\ \nu_0 - threshold\ frequency\ \phi_0 - work function$

- wherein



Maximum kinetic energy of emitted electron is:

$$K_{max} = \frac{hc}{\lambda} - \phi = \left(\frac{1240}{500} - 2.28\right) ev$$

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$= (2.48 - 2.38)ev = 0.10\,ev$

de Broglie wavelength of the emitted electron will be:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(K.E)}}$$

$$\lambda = \frac{6.67 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-20}}}$$

$$= \frac{6.67 \times 10^{-34}}{\sqrt{2.912 \times 10^{-50}}} = \frac{6.67 \times 10^{-9}}{1.70}$$

$$\lambda = 3.9 \times 10^{-9} m \geqslant 2.8 \times 10^{-9} m$$

Q. 12 Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is :

Option 1:
$\frac{\sigma_1 + \sigma_2}{2\sigma_1\sigma_2}$

Option 2:
$\frac{\sigma_1 + \sigma_2}{\sigma_1\sigma_2}$

Option 3:
$\frac{\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$

Option 4:
$\frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$

Correct Answer:
$\frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}$

Solution:

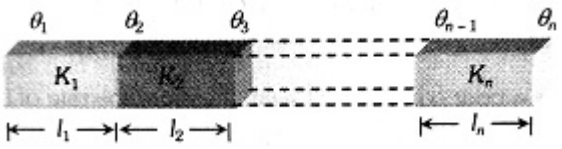
In Series Combination -

$$R_{eq} = R_1 + R_2 + R_3 - - - - - - - - - - R_n$$

- wherein

R_{eq}= equivalent thermal resistance

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In series combination

Resistance $R_{eq} = R_1 + R_2$

$$R = \frac{L}{KA} \Rightarrow R_1 = \frac{L}{\sigma_1 A}$$

$$R_2 = \frac{L}{\sigma_2 A}$$

$$R_{eq} = \frac{2L}{\sigma_{eq} A} = \frac{L}{\sigma_1 A} + \frac{L}{\sigma_1 A}$$

OR $= \frac{2}{\sigma_{eq}} = \frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

OR $\sigma_{eq} = \frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$

Q. 13 An automobile moves on a road with a speed of 54 km h^{-1}. The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kg m^2. If the vehicle is brought to rest in 15 s, the magnitude of average torque transmitted by its brakes to the wheel is :

Option 1:
8.58 kg m^2 s^{-2}

Option 2:
10.86 kg m^2 s^{-2}

Option 3:
2.86 kg m^2 s^{-2}

Option 4:
6.66 kg m^2 s^{-2}

Correct Answer:
6.66 kg m^2 s^{-2}

Solution:
As we learnt in

Analogue of second law of motion for pure rotation -

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$\vec{\tau} = I\,\alpha$

- wherein

Torque equation can be applied only about two point

(i) centre of motion.

(ii) point which has zero velocity/acceleration.

V=54 Km/hr = 15m/s

Angular velocity of vehicle $V=W_0 r$

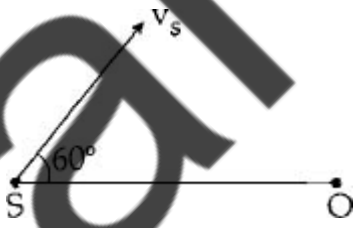
$$W_0 = \frac{V}{R} = \frac{15}{0.45} = \frac{100}{3} rad/s$$

Angular Acc^n $$\propto = \frac{\Delta W}{t} = \frac{-100}{45}\, rad/s^2$$

$$\vec{\tau} = I \propto$$

$$\Rightarrow 3 \times \frac{100}{45} = 6.66\ Kg\, m^2 s^{-2}$$

Q. 14 A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}), is:



Option 1:
103 Hz

Option 2:
106 Hz

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Option 3:
97 Hz

Option 4:
100 Hz

Correct Answer:
103 Hz

Solution:
From Doppler's effect,

$$\nu' = \nu_0 \left(\frac{v - v_0}{v - v_s} \right)$$

$$\nu' = 100 \left(\frac{v - 0}{v - (+9.7)} \right)$$

$$\Rightarrow \nu' = \frac{100v}{v\left(1 - \frac{9.7}{v}\right)}$$

$$= \frac{100}{\left(1 - \frac{9.7}{330}\right)}$$

$$\nu' = 103Hz$$

The correct option is 1.

Q. 15 On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision is :

Option 1:
$\frac{3}{4}v$

Option 2:
$\frac{3}{\sqrt{2}}v$

Option 3:
$\frac{\sqrt{3}}{2}v$

Option 4:
$\frac{2\sqrt{2}}{3}v$

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Correct Answer:

$\frac{2\sqrt{2}}{3}v$

Solution:

As we learnt in

Conservation of Energy and Conservation of Mechanical Energy -

Accounting all forms of energy within isolated system

i.e. total energy remains constant.

Perfectly Elastic Collision -

Law of conservation of momentum and that of Kinetic Energy hold good.

- wherein

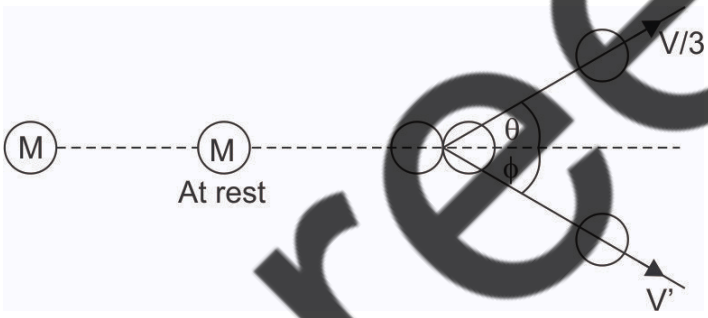
$\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2=\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2$

$m_1u_1+m_2u_2=m_1v_1+m_2v_2$

$m_1, m_2 : masses$

$u_1, v_1 : initial\ and\ final\ velocity\ of\ the\ mass\ m_1$

$u_2, v_2 : initial\ and\ final\ velocity\ of\ the\ mass\ m_2$



According to conversation of energy

$\frac{1}{2}Mv^2+0=\frac{1}{2}M\left(\frac{v}{3}\right)^2+\frac{1}{2}Mv'^2$

$v^2=\frac{v^2}{9}+v'^2$

$\Rightarrow v'^2=v^2-\frac{v^2}{9}$

$=\frac{8}{9}v^2$

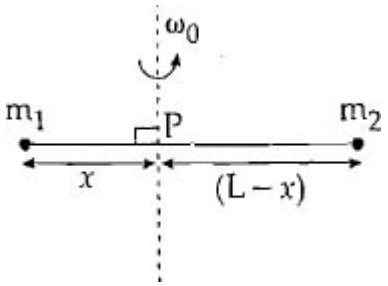
$\Rightarrow v'^2=\frac{8}{9}v^2$

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$$v' = \frac{2\sqrt{2}}{3}v$$

Correct option is 4.

Q. 16 Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L, and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by :



Option 1:

$$x = \frac{m_1}{m_2}L$$

Option 2:

$$x = \frac{m_2}{m_1}L$$

Option 3:

$$x = \frac{m_2 L}{m_1 + m_2}$$

Option 4:

$$x = \frac{m_1 L}{m_1 + m_2}$$

Correct Answer:

$$x = \frac{m_2 L}{m_1 + m_2}$$

Solution:

As we learnt in

Moment of inertia for system of particle -

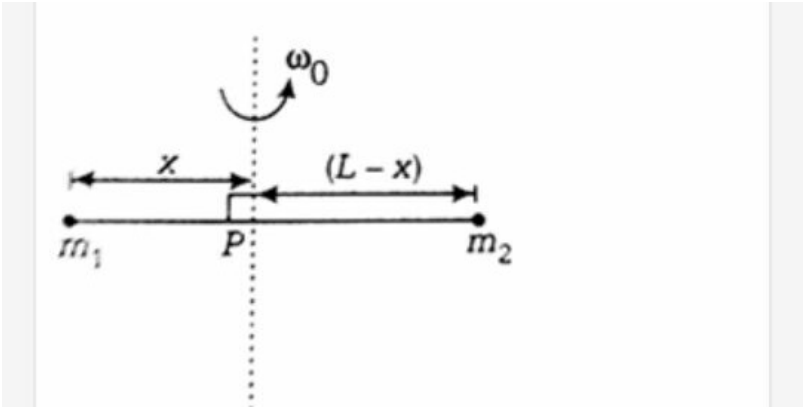
$$I = m_1 r_1^2 + m_2 r_2^2 + ...m_n r_n^2$$

$$= \sum_{i=1}^{n} m_i r_i^2$$

- wherein

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Applied when masses are placed discretely.



Total M.I of rod

$$I = m_1x^2 + m_2(L-x)^2$$

$$I = m_1x^2 + m_2L^2 + m_2x^2 - m_2Lx$$

As I is minimum i.e.

$$\frac{di}{dx} = 2m_1x + 0 + 2xm_2 - 2m_2L = 0$$

$$=> x(2m_1 + 2m_2) = 2m_2L$$

$$x = \frac{m_2L}{m_1 + m_2}$$

Q. 17 A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, loses 50 percent of its energy in collision and rebounds to the same height. The initial velocity v_0 is : (Take g = 10 ms^{-2})

Option 1:
20 ms^{-1}

Option 2:
28 ms^{-1}

Option 3:
10 ms^{-1}

Option 4:
14ms^{-1}

Correct Answer:
20 ms^{-1}

Solution:
As we learnt in

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Kinetic energy -

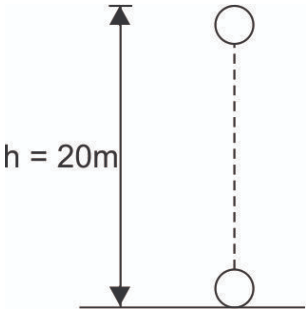
$$k = \frac{1}{2}mv^2$$

- wherein

$m \rightarrow mass$

$v \rightarrow velocity$

kinetic Energy is never negative



When balls collide with the ground it loses its's 50% of energy

$$\therefore \frac{K_f}{K_i} = \frac{1}{2}$$

$$= \frac{\frac{1}{2}mv_f^2}{\frac{1}{2}mv_i^2}$$

$$\frac{v_f}{v_i} = \frac{1}{\sqrt{2}}$$

$$=> \frac{\sqrt{2gh}}{\sqrt{u^2 + 2gh}} = \frac{1}{\sqrt{2}}$$

$$4gh = u^2 + 2gh$$

$$u = 20ms^{-1}$$

Q. 18 A nucleus of uranium decays at rest into nuclei of thorium and helium. Then:

Option 1:
The helium nucleus has less momentum than the thorium nucleus.

Option 2:
The helium nucleus has more momentum than the thorium nucleus.

Option 3:
The helium nucleus has less kinetic energy than the thorium nucleus.

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Option 4:
The helium nucleus has more kinetic energy than the thorium nucleus.

Correct Answer:
The helium nucleus has more kinetic energy than the thorium nucleus.

Solution:
As we learnt in

? -decay -

$$^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + ^{4}_{2}He + Q$$

- wherein

$$Q\ value = (M_X - M_Y - M_{He})\,c^2$$

$$\bigcup_{92}^{238} \rightarrow ^{234}_{90}\mathrm{Th} + ^{4}_{2}\mathrm{He}$$

Since from conservation of momentum, momentum of Thorium and Helium is same. Hence kinetic energy of Helium will be greater than kinetic energy of thorium.

$$|\vec{p}_{Th}| = |\vec{p}_{He}| = p$$

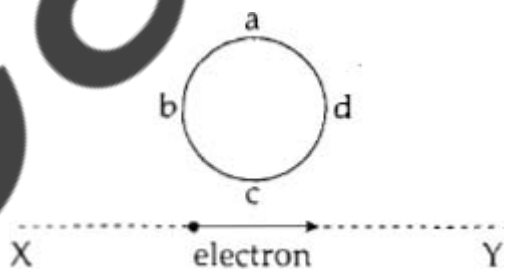
$$K.E._{Th} = \frac{p^2}{2m_{Th}};\ \ K.E._{Th} = \frac{p^2}{2m_{He}}$$

$$\because m_{He} < m_{Th}$$

$$\therefore K.E_{He} > K.E_{Th}$$

Correct option is 4.

Q. 19 An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil ?



Option 1:
adcb

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Option 2:
The current will reverse its direction as the electron goes past the coil

Option 3:
No current induced

Option 4:
abcd

Correct Answer:
The current will reverse its direction as the electron goes past the coil

Solution:
When the electron moves from X to Y. the flux linked with the coll abcd (which is into the page) will first increase and then decrease as the electron passes by. So the induced current in the coil will be first anticlockwise and will reverse its direction (i.e. will become clockwise) as the electron goes past the coil.

The correct option is 2.

Q. 20 A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be :

Option 1:
$\frac{\alpha}{\beta}$

Option 2:
$\frac{\beta^2}{\alpha}$

Option 3:
$\frac{2\pi\beta}{\alpha}$

Option 4:
$\frac{\beta^2}{\alpha^2}$

Correct Answer:
$\frac{2\pi\beta}{\alpha}$

Solution:
$x = A\sin(wt + \delta)$

$\alpha = +w^2A$ (Maximum acceleration)

$\beta = Aw$ (Maximum velocity)

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$$w = \frac{\alpha}{\beta}$$

$$w = \frac{2\pi}{T} \quad \Rightarrow T = \frac{2\pi}{w} = \frac{2\pi\beta}{\alpha}$$

Hence, the answer is the option (3).

Q. 21 Two slits in Youngs experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is :

Option 1:
$\frac{121}{49}$

Option 2:
$\frac{49}{121}$

Option 3:
$\frac{4}{9}$

Option 4:
$(\frac{6}{4})^2$

Correct Answer:
$(\frac{6}{4})^2$

Solution:

$$\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2$$

$$I_2 = 25I, since \frac{I_2}{I_1} = \frac{W_2}{W_1} = \frac{25}{1}$$

$$\therefore \frac{Imax}{Imin} = \left(\frac{\sqrt{I_1}+\sqrt{25I_1}}{\sqrt{I_1}-\sqrt{25I_1}}\right)^2 = \left(\frac{6}{4}\right)^2$$

Hence, the answer is the option (4).

Q. 22 If potential (in volts) in a region is expressed as *V(x, y, z) = 6xy - y + 2yz,* the electric field (in N/C) at point (1, 1, 0) is :

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Option 1:
$-\left(6\hat{i}+5\hat{j}+2\hat{k}\right)$

Option 2:
$-\left(2\hat{i}+3\hat{j}+\hat{k}\right)$

Option 3:
$-\left(6\hat{i}+9\hat{j}+\hat{k}\right)$

Option 4:
$-\left(3\hat{i}+5\hat{j}+3\hat{k}\right)$

Correct Answer:
$-\left(6\hat{i}+5\hat{j}+2\hat{k}\right)$

Solution:

$$E=-\left[\frac{dV}{dx}\hat{j}+\frac{dV}{dy}\hat{j}+\frac{dV}{dz}\hat{k}\right]$$

$$V(x,y,z)=6xy-y+2yz$$

$$\therefore E=-\left[\frac{d}{dx}\left(6xy-y+2yz\right)\hat{i}+\frac{d}{dy}\left(6xy-y+2yz\right)\hat{j}+\frac{d}{dz}\left(6xy-y+2yz\right)\hat{k}\right]$$

At point $(1,1,0)$

$$\vec{E}=-\left[6\left(1\right)\hat{i}+6\left(1\right)-1+2\left(0\right)\hat{j}+2\left(1\right)\hat{k}\right]$$

Hence, the answer is the option (1).

Q. 23 A parallel plate air capacitor has capacity 'C, distance of separation between plates is 'd' and potential difference 'V' is applied between the plates. Force of attraction between the plates of the parallel plate air capacitor is :

Option 1:
$\frac{\mathrm{CV}^2}{2\ \mathrm{d}}$

Option 2:
$\frac{\mathrm{CV}^2}{\mathrm{d}}$

Option 3:
$\frac{\mathrm{C}^2\mathrm{V}^2}{2\ \mathrm{d}^2}$

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Option 4:

$$\frac{C^2V^2}{2\ d}$$

Correct Answer:

$$\frac{CV^2}{2\ d}$$

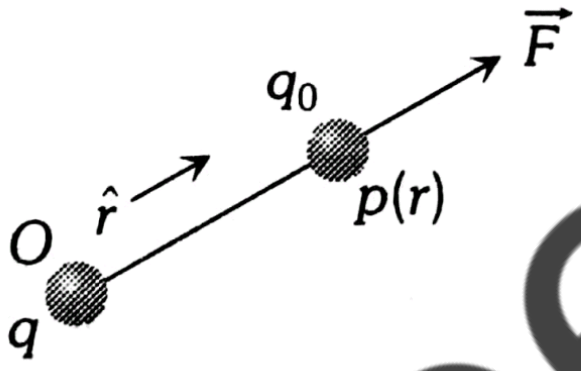
Solution:

As we learnt in

Electric Field Intensity -

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{kQ}{r^2}$$

- wherein



AND

Parallel Plate Capacitor -

$$C = \frac{\epsilon_0 A}{d}$$

- wherein

Area - A seperation between two plates.

Force between the pilates

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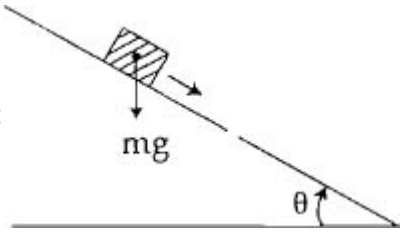
$$F = qE \;\; = q \times \frac{\sigma}{2\varepsilon_o} = q \times \frac{q}{2A\varepsilon_o}$$

$$F = \frac{q^2}{2\varepsilon_o A}$$

$$q = cv \;\; \Rightarrow \;\; c = \frac{\varepsilon_o A}{d} \;\; \Rightarrow \varepsilon_o A = cd$$

$$F = \frac{c^2 v^2}{2cd} = \frac{cv^2}{2d}$$

Q. 24 A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30°, the box starts to slip and slides 4.0 m down the plank in 4.0 s. The coefficients of static and kinetic friction between the box and the plank will be, respectively:



Option 1:
0.6 and 0.5

Option 2:
0.5 and 0.6

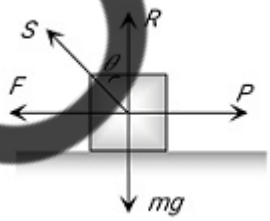
Option 3:
0.4 and 0.3

Option 4:
0.6 and 0.6

Correct Answer:
0.6 and 0.5

Solution:
As we learnt in

Angle of Friction -



Angle θ is called angle of friction.

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$tan\theta = \frac{F_l}{R}$

$tan\theta = \mu_s$

R = Reaction, $\frac{F_l}{R} = \mu_s$

$F_l =$ Force of limiting friction

$\theta = tan^{-1}(\mu_s)$

- wherein

$\mu_s =$ static friction coefficient

$\theta =$ Angle made by resultant of limiting friction and R with the normal reaction.

$$M_s = tan\ 30^\circ = \frac{1}{\sqrt{3}} = 0.6$$

$$S = ut + \frac{1}{2}at^2, \quad u = 0, \quad a = g(sin\theta - Mcos\theta)$$

$$4 = \frac{1}{2}g(sin30 - M_k cos30)4^2$$

$$\Rightarrow 0.5 \times 10 \times \frac{1}{2} - M_k \times 10 \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow 5\sqrt{3}M_k = 4.5 \quad \Rightarrow M_k = 0.51$$

Correct option is 1.

Q. 25 In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is :

Option 1:

$\frac{9}{4}$

Option 2:

$\frac{27}{5}$

Option 3:

$\frac{5}{27}$

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Option 4:

$\frac{4}{9}$

Correct Answer:

$\frac{5}{27}$

Solution:

The longest wavelength in the Balmer series corresponds to the transition between, Let's say it is λ_1

$$3 \rightarrow 2 \ \ then \ \frac{1}{\lambda_1} = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5}{36}R$$

The longest wavelength in the layman series corresponds to the transition between $2 \rightarrow 1$, Let's say it is λ_2

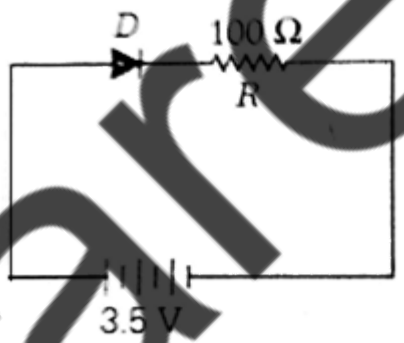
Then,

$$\frac{1}{\lambda_2} = R\left(1 - \frac{1}{4}\right) = \frac{3R}{4}$$

$$\frac{\lambda_{lymen}}{\lambda_{Balmer}} = \frac{\frac{4}{3R}}{\frac{36}{5R}} = \frac{4}{3R} \times \frac{5R}{36} = \frac{5}{27}$$

Hence, the answer is the option (3).

Q. 26 In the given figure, a diode D is connected to an external resistance R = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be :



Option 1:

40 mA

Option 2:

20 mA

Option 3:

35 mA

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Option 4:
30 mA

Correct Answer:
30 mA

Solution:
As we learnt in

P -N junction as diode -

It is a one way device. It offers a low resistance when forward biased and high resistance when reverse biased.

- wherein

R = 0, Forward

R $\rightarrow \infty$ Reverse

The emf developed across diode is 0.5 V.

Total applies potential = 3.5 V

$\therefore$ Potential difference across resistance of 100 Ω is 3 V.

Current across 100 Ω resistance $\frac{3\ V}{100\Omega} = 30\ mA$

Q. 27 A satellite *S* is moving in an elliptical orbit around the Earth. The mass of the satellite is very small compared to the mass of the Earth. Then,

Option 1:
the total mechanical energy of *S* varies periodically with time.

Option 2:
the linear momentum of S remains constant in magnitude.

Option 3:
the acceleration of S is always directed towards the centre of the earth.

Option 4:
the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant.

Correct Answer:
the acceleration of S is always directed towards the centre of the earth.

Solution:

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As $h = 0$ and $GM = gR^2$

The motion of a satellite is due to force applied by the earth on the satellite and hence its acceleration is always towards the center of the earth. Angular momentum and total energy of the satellite are conserved. Linear momentum is not constant.

Hence, the answer is the option (3).

Q. 28 A force $\vec{\mathrm{F}} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{\mathrm{r}} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value α for which angular momentum about the origin is conserved is:

Option 1:
2

Option 2:
zero

Option 3:
1

Option 4:
-1

Correct Answer:
-1

Solution:

$$L = r^{\rightarrow} \times p^{\rightarrow} = \ constant$$

$$L = \begin{vmatrix} 2 & -6 & -12 \\ \alpha & 3 & 6 \end{vmatrix} = constant$$

On solving, we get:

$$\alpha = -1$$

Hence, the answer is the option (4).

Q. 29 A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length *l* of the potentiometer wire. The e.m.f. E will be given by:

Option 1:

$$\frac{\mathrm{E_0 r}}{(\mathrm{r} + \mathrm{r_1})} \cdot \frac{l}{\mathrm{L}}$$

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Option 2:
$\frac{\mathrm{E}_0 l}{\mathrm{L}}$

Option 3:
$\frac{\mathrm{LE}_0\ \mathrm{r}}{(\mathrm{r}+\mathrm{r}_1)\, l}$

Option 4:
$\frac{\mathrm{LE}_0\ \mathrm{r}}{l\ \mathrm{r}_1}$

Correct Answer:
$\frac{\mathrm{E}_0 \mathrm{r}}{(\mathrm{r}+\mathrm{r}_1)} \cdot \frac{l}{\mathrm{L}}$

Solution:
Current in the circuit having E_0 is: $i = \frac{E_0}{r+\eta_1}$

voltage dropped in potentiometer wire of length L is: $ir = \frac{E_0 r}{r + r_1}$

since the balanced point is obtained at length I of the potentiometer wire, the emf of the cell E has to match the voltage dropped in length I of the wire.

$$\therefore E = \frac{E_0 r}{r + r_1} \times \left(\frac{l}{L}\right)$$

Q. 30 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 $JK^{-1}\ mol^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1}, then the heat capacity at constant pressure is

(Take gas constant R = 8.3 $JK^{-1}\ mol^{-1}$)

Option 1:
7.5 $JK^{-1}\ mol^{-1}$

Option 2:
7.0 $JK^{-1}\ mol^{-1}$

Option 3:
8.5 $JK^{-1}\ mol^{-1}$

Option 4:
8.0 $JK^{-1}\ mol^{-1}$

Correct Answer:
8.0 $JK^{-1}\ mol^{-1}$

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Solution:
As we learnt in

Atomicity or adiabatic coefficient (gamma) -

$$\gamma = \frac{C_p}{C_v}$$

$$= 1 + \frac{2}{f}$$

- wherein

for Monoatomic gas $\gamma = \frac{5}{3}$

for Diatomic gas $\gamma = \frac{7}{5}$

for Triatomic gas $\gamma = \frac{4}{3}$

given M = 4gm

$$V = 22.4L$$

$$C_v = 5.0JK^{-1}mol^{-1}$$

$$V_{sound} = 952m/s$$

$$C_p = ?$$

Velocity of sound

$$V_{sound} = \sqrt{\frac{rpV}{M}}$$

$$or\ r = \frac{MV_{sound}^2}{PV} = \frac{(952)^2 \times 4 \times 10^{-3}}{10^5 \times 22.4 \times 10^{-3}}$$

$$\because r = \frac{C_p}{C_v} => C_p = rc_v = \frac{(952)^2 \times 4 \times 10^{-3} \times 5}{10^5 \times 22.4 \times 10^{-3}}$$

$$=> C_p = 8.09J/mol - K$$

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Q. 31 Two stones of masses m and 2m are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and the lighter one in radius r. The tangential speed of lighter stone is n times that of the value of heavier stone when they experience the same centripetal forces. The value of n is:

Option 1:
3

Option 2:
4

Option 3:
1

Option 4:
2

Correct Answer:
2

Solution:
F_c (heavy)= F_c(light)

$$\frac{2m(v)^2}{r/2} = \frac{m(nv)^2}{2}$$

$n^2 = 4$

Or, n = 2

Hence, the answer is the option (4).

Q. 32 A remote-sensing satellite of Earth revolves in a circular orbit at a height of 0.25 x 10^6 m above the surface of Earth. If Earth's radius is 6.38 x 10^6 m and g = 9.8 ms^{-2}, then the orbital speed of the satellite is:

Option 1:
8.56 km s^{-1}

Option 2:
9.13 km s^{-1}

Option 3:
6.67 km s^{-1}

Option 4:
7.76 km s^{-1}

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Correct Answer:
7.76 km s^{-1}

Solution:
The orbital velocity of the satellite is

$v = \sqrt{\frac{GM}{R+h}}$ where M= mass of the earth and R= radius of the earth.

so,

$$v = \sqrt{\frac{6.67 \times 10^{-11} \times (6 \times 10^{24})}{(6.4 \times 10^{6}) + (0.25 \times 10^{6})}} = 7.76\text{km/s}$$

Hence, the answer is the option (4).

Q. 33 A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is:

Option 1:
205 Hz

Option 2:
10.5 Hz

Option 3:
105 Hz

Option 4:
155 Hz

Correct Answer:
105 Hz

Solution:
As we leant in

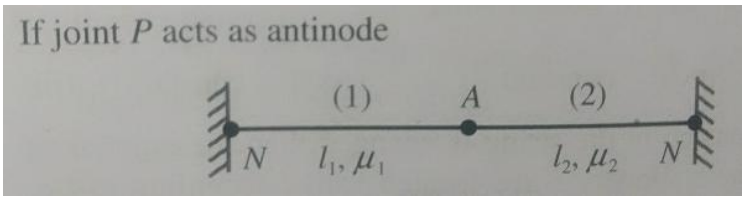
Fundamental frequency with end correction -

$\nu_0 = \frac{V}{4\,(l+e)}$ (one end open)

$\nu_0 = \frac{V}{2\,(l+2e)}$ (Both end open)

e = end correction

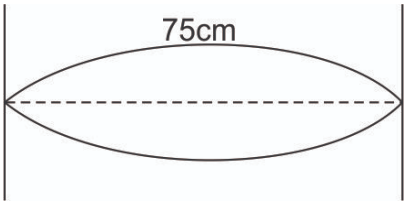
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-

In a stretched string all multiples of frequencies can be obtained, i.e., If fundamental frequency is η, then higher frequencies will be $2\eta, 3\eta, 4\eta$ .....

According to question,



$$\eta = 420 - 315 = 105\ Hz$$

Correct option is 3.

Q. 34 The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is -20°C, the temperature of the surroundings to which it rejects heat is:

Option 1:
41°C

Option 2:
11°C

Option 3:
21°C

Option 4:
31°C

Correct Answer:
31°C

Solution:
As we learnt in

Coefficient of performance (beta) -

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$$

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- wherein

$T_1 > T_2$

For a perfect refrigerator $\beta \to \infty$

Coefficient of performance of a refrigerator is

$$\beta = \frac{T_2}{T_1 - T_2} \quad .....(1)$$

$T_2 = -20^\circ C = 253K$

β = 5

From equation (1)

$$5 = \frac{253}{T_1 - 253}$$

or $5T_1 - 1265 = 253$

or $5T_1 = 1518$

$T_1 = 303.6K = 30.6^\circ C$

Correct option is 4.

Q. 35 Water rises to a height of 'h' in a capillary tube. If the length of the capillary tube above the surface of the water is made less than 'h', then :

Option 1:
water rises upto the top of capillary tube and stays there without overflowing.

Option 2:
water rises upto a point a little below the top and stays there.

Option 3:
water does not rise at all.

Option 4:
water rises upto the tip of capillary tube and then starts overflowing like a fountain.

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Correct Answer:
water rises upto the top of capillary tube and stays there without overflowing.

Solution:
Ascent Formula -

$$\frac{2T\cos\Theta}{\rho g r}$$

If the length of the capillary tube is insufficient then the contact angle will adjust itself such that water rises to the top of the capillary tube and stays there without flowing.

Hence, the answer is the option (1).

Q. 36 Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is:

Option 1:
$\frac{3}{4}$

Option 2:
2

Option 3:
$\frac{1}{2}$

Option 4:
$\frac{2}{3}$

Correct Answer:
$\frac{3}{4}$

Solution:
Ideal gas equation PV=nRT

OR $PM = \rho RT$

where $M = Molecular\ weigh$

$\rho = density$

$For\ A = P_A.M_A = \rho_A RT$

$For\ B = P_B.M_B = \rho_B RT$

Divide the two equation

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$$\frac{P_A M_A}{P_B M_B} = \frac{\rho_A}{\rho_B} .....(1)$$

Given $P_A = 2P_B$, $\rho_A = 1.5\rho_B$

$$\therefore \frac{M_A}{M_B} = (\frac{\rho_A}{\rho_B})(\frac{\rho_B}{\rho_A}) = (1.5)(\frac{1}{2}) = \frac{3}{4}$$

Hence, the answer is the option (1).

Q. 37 The Young's modulus of steel is twice that of brass. Two wires of the same length and same area of cross-section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of:

Option 1:
2:1

Option 2:
4:1

Option 3:
1:1

Option 4:
1:2

Correct Answer:
2:1

Solution:
Given,

$$Y_{steel} = 2Y_{brass}$$

$$l_{steel} = l_{brass}$$

$$A_{steel} = A_{brass}$$

$$Y = \frac{Fl}{A\,\Delta l} \quad or \quad \Delta l = \frac{Fl}{AY}$$

According to the question,

$$\Delta l_{steel} = \Delta l_{brass} \quad \Rightarrow \left(\frac{F}{Y}\right)_{steel} = \left(\frac{F}{Y}\right)_{brass}$$

$$\frac{F_{steel}}{F_{brass}} = \frac{Y_{steel}}{Y_{brass}} = 2:1$$

Hence, the answer is the option (1).

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Q. 38 The input signal given to a CE amplifier having a voltage gain of 150 is $V_i = 2cos\left(15t + \frac{\pi}{3}\right)$. The corresponding output signal will be :

Option 1:

$75\cos\left(15t + \frac{2\pi}{3}\right)$

Option 2:

$2\cos\left(15\,t + \frac{5\pi}{6}\right)$

Option 3:

$300\cos\left(15\,t + \frac{4\pi}{3}\right)$

Option 4:

$300\cos\left(15\,t + \frac{\pi}{3}\right)$

Correct Answer:

$300\cos\left(15\,t + \frac{4\pi}{3}\right)$

Solution:

As we learnt in

Amplifier -

Transistor is used as an amplifier

- wherein

in such circuit output is amplified as compared to input .

Input signal of a CE amplifier, $V_i = 2\cos\left(15\,t + \frac{\pi}{3}\right)$

Voltage gain = 150

As CE amplifier give phase difference of π between input and output signals,

so $A_v = \frac{V_0}{V_i} \Rightarrow V_0 = A_v \cdot V_i$

$V_0 = 150 \times 2\cos\left(15\,t + \frac{\pi}{3} + \pi\right)$

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$$V_0 = 300 \cos\left(15\,t + \frac{4\pi}{3}\right)$$

Q. 39 In an astronomical telescope in normal adjustment a straight black line of length L is drawn on inside part of objective lens. The eye-piece forms a real image of this line. The length of this image is I. The magnification of the telescope is :

Option 1:
$\frac{\mathrm{L}}{\mathrm{I}} - 1$

Option 2:
$\frac{\mathrm{L}+\mathrm{I}}{\mathrm{L}-\mathrm{I}}$

Option 3:
$\frac{\mathrm{L}}{\mathrm{I}}$

Option 4:
$\frac{\mathrm{L}}{\mathrm{I}} + 1$

Correct Answer:
$\frac{\mathrm{L}}{\mathrm{I}}$

Solution:
As we learnt in

Length of Astronomical Telescope -

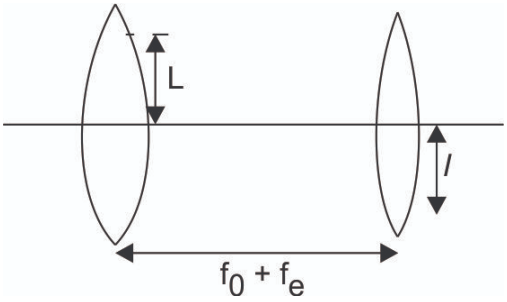
$$L = f_o + \frac{f_e D}{f_e + D}$$

- wherein

f_o = Focal length of objective.

f_e = Focal length of eyepiece

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Magnification of telescope is

$$M = \frac{fo}{fe}$$

Here

$$\frac{fe}{fe+u} = \frac{-l}{L}$$

$$=> \frac{fe}{fe-(fo+fe)} = \frac{-l}{L} \ (Since\ u = -fo + fe)$$

$$or \frac{-fe}{fo} = \frac{-l}{L}$$

$$or\ \frac{fe}{fo} = \frac{l}{L}$$

$$=> M = \frac{fo}{fe} = \frac{L}{l}$$

Correct option is 3.

Q. 40 The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be 13.6 x 10^3 kg/m^3 and g = 10 m/s^2 then the power of heart in watt is :

Option 1:
2.35

Option 2:
3.0

Option 3:
1.50

Option 4:
1.70

Correct Answer:
1.70

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Solution:
As we learnt in Pressure gradient -

$$\frac{dp}{dh} = \rho g$$

$\therefore$ Pressure varies with respect to depth / height

Volume rate of flow=$\frac{5 \times 10^{-3}}{60}\ m^3/sec$

Pressure=150 mm of Hg

$$\rho g h = 13.6 \times 10^3 \times 10 \times 150 \times 10^{-3} = 20400\ N/m^2$$

Power=$\frac{dW}{dt} = \frac{d}{dt}(P\Delta V) = P \cdot \frac{d}{dt}(\Delta V)$

$$= \frac{20400 \times 5 \times 10^{-3}}{60} W = 1.7W$$

Hence, the answer is the option(4).

Q. 41 If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $[\eta^x \rho^y r^z]$, where η, ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of *x*, *y* and *z* are given by :

Option 1:
-1, -1, 1

Option 2:
-1, -1, -1

Option 3:
1, 1, 1

Option 4:
1, -1, -1

Correct Answer:
1, -1, -1

Solution:
- Coefficient of viscosity (η) : $[\eta] = \left[M^1 L^{-1} T^{-1}\right]$
- Density (ρ) : $[\rho] = \left[M^1 L^{-3} T^0\right]$
- Radius(r) : $[r] = \left[L^1\right]$
- Critical velocity (V_c) : $[V_c] = \left[M^0 L^1 T^{-1}\right]$

Given equation: $V_c = \eta^x \rho^y r^z$
Express the dimensions of each variable in terms of its powers:

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$$\left[M^0L^1T^{-1}\right] = \left[M^xL^{-x-3y+z}T^{-x}\right]$$

Compare powers of M, L, and T on both sides:
- Equation $1: x + y = 0$
- Equation 2: $-x - 3y + z = 1$

From equation (3), solve for x :

$$-x = -1 \Longrightarrow x = 1$$

Substitute $x = 1$ into equation (1):

$$1 + y = 0 \Longrightarrow y = -1$$

Substitute $x = 1$ and $y = -1$ into equation (2) to solve for z :

$$-1 - 3(-1) + z = 1 \Longrightarrow -1 + 3 + z = 1 \Longrightarrow z = -1$$

So, the values of x, y, and z are $x = 1, y = -1$, and $z = -1$ respectively.
Hence, the answer is the option (4).

Q. 42 A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Planck's constant, c = speed of light)

Option 1:
$\frac{\mathrm{hc}}{\lambda}$

Option 2:
$\frac{2\ \mathrm{hc}}{\lambda}$

Option 3:
$\frac{\mathrm{hc}}{3\lambda}$

Option 4:
$\frac{\mathrm{hc}}{2\lambda}$

Correct Answer:
$\frac{\mathrm{hc}}{2\lambda}$

Solution:
As we learnt in

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Conservation of energy -

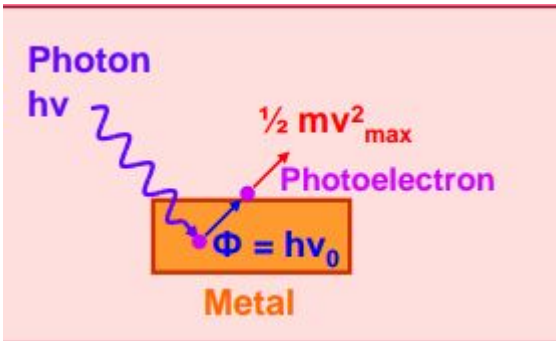
$$h\nu = \phi_0 + \frac{1}{2}mv_{max}^2$$

$$h\nu = h\nu_0 + \frac{1}{2}mv_{max}^2$$

$$h\left(\nu - \nu_0\right) = \frac{1}{2}mv_{max}^2$$

$where, h - Plank's constant\ \nu - Frequency\ \nu_0 - threshold\ frequency\ \phi_0 - work function$

- wherein



Let work function be ϕ

$$K_1 = K.E_{max} = \frac{hc}{\lambda} - \phi \qquad -(i)$$

$$K_2 = \frac{hc}{\frac{\lambda}{2}} - \phi = 2\frac{hc}{\lambda} - \phi \qquad -(ii)$$

According to the question,

$$\frac{2hc}{\lambda} - \phi = 3\left(\frac{hc}{\lambda} - \phi\right)$$

Or, $\frac{2hc}{\lambda} - \phi = \frac{3hc}{\lambda} - 3\phi$

Or, $\phi = \frac{hc}{2\lambda}$

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Q. 43 The cylindrical tube of a spray pump has a radius R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is :

Option 1:
$\frac{VR^2}{nr^2}$

Option 2:
$\frac{VR^2}{n^3r^2}$

Option 3:
$\frac{V^2R}{nr}$

Option 4:
$\frac{VR^2}{n^2r^2}$

Correct Answer:
$\frac{VR^2}{nr^2}$

Solution:
Mass of the liquid entering per second at A = mass of the liquid leaving per second at B.

$a_1 v_1 = a_2 v_2$

wherein, a_1 and a_2 be the area of the cross-section.

Consider a cylinder tube of a spray pump that has radius R_1 one end having n fine holes each of radius r and the speed of liquid in the tube is v.

$\because Av = constant$

$\Rightarrow \pi R^2 v = n.\pi r^2 v'$

$\Rightarrow v' = \frac{v.R^2}{nr^2}$

Hence, the answer is the option (1).

Q. 44 If vectors $\vec{A} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ and $\vec{B} = \cos\frac{\omega t}{2}\,\hat{i} + \sin\frac{\omega t}{2}\,\hat{j}$ are functions or time, then the value of t at which they are orthogonal to each other is :

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Option 1:
$t = \frac{\pi}{2\omega}$

Option 2:
$t = \frac{\pi}{\omega}$

Option 3:
$t = 0$

Option 4:
$t = \frac{\pi}{4\omega}$

Correct Answer:
$t = \frac{\pi}{\omega}$

Solution:
As we learnt in

Composition of two SHM in perpendicular direction -

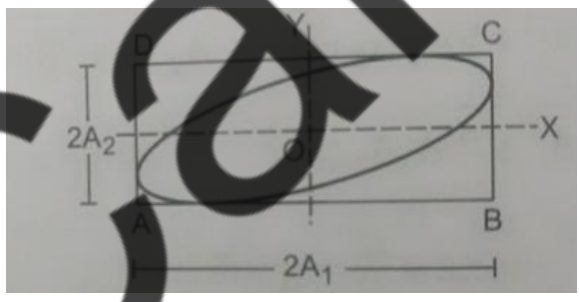
$x = A_1 \sin \omega t$

$y = A_2 \sin (\omega t + \delta)$

- wherein

Resultant equation

$$\frac{x^2}{{A_1}^2} + \frac{{y_2}^2}{{A_2}^2} - \frac{2xy \cos \delta}{A_1 A_2} = \sin^2 \delta$$



$$\vec{A}.\vec{B} = (\cos \omega t \hat{i} + \sin \omega t \hat{j}).\left(\cos \frac{\omega t}{2}\,\hat{i} + \sin \frac{\omega t}{2}\,\hat{j}\right) = \cos\left(\omega t - \frac{\omega t}{2}\right)$$

When they are orthogonal, $\vec{A}.\vec{B} = 0$

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$$\cos\left(\omega t - \frac{\omega t}{2}\right) = 0$$

$$\cos\left(\frac{\omega t}{2}\right) = 0 \qquad \Rightarrow \frac{\omega t}{2} = \frac{\pi}{2}$$

$$t = \frac{\pi}{\omega}$$

Correct option is 2.

Q. 45 A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m^2. The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be :

Option 1:
0.20 Nm

Option 2:
0.24 Nm

Option 3:
0.12 Nm

Option 4:
0.15 Nm

Correct Answer:
0.20 Nm

Solution:
The torque on a loop in a uniform magnetic field is given by :

$$\vec{\tau} = [\vec{M} \times \vec{B}]\vec{\tau}| = |\vec{M}||\vec{B}|\sin\theta]$$

$where, \theta$ is the angle between the normal and the plane of the loop.

$$\vec{M} = NI \times l \times b = 50 \times 2 \times 0.12 \times 0.1 = 1.2]$$

$$\vec{B} = 0.2\,\mathrm{T}\vec{\tau} = 1.2 \times 0.2 \times \sin 60^\circ$$

$$\vec{\tau} = 0.20\,\mathrm{Nm}$$

Hence, the answer is the option(1).

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Chemistry

Q. 1 Reaction of phenol with chloroform in presence of dilute sodium hydroxide finally introduces which one of the following functional group ?

Option 1:
$-CH_2Cl$

Option 2:
-COOH

Option 3:
$-CHCl_2$

Option 4:
-CHO

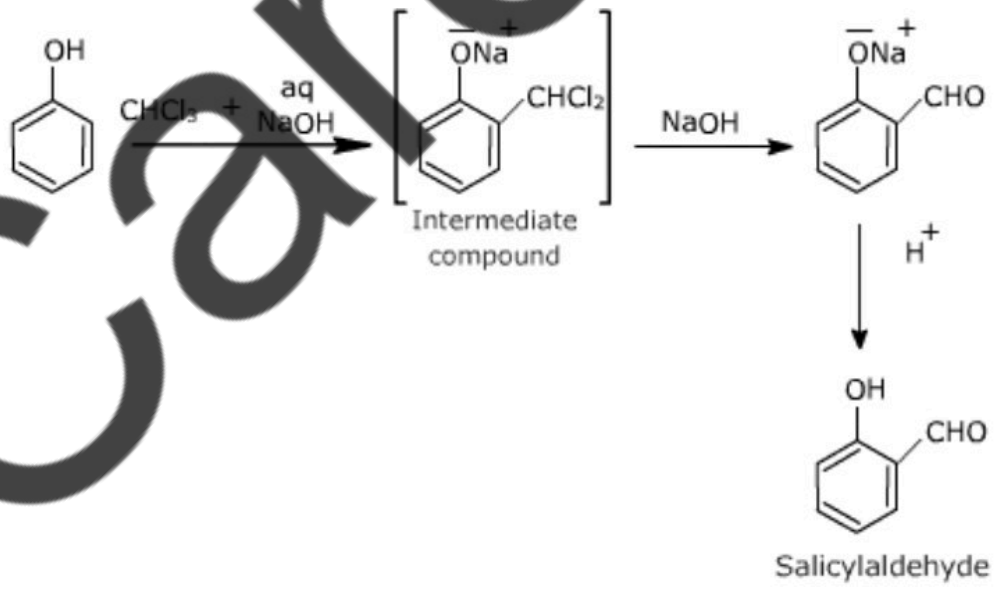
Correct Answer:
-CHO

Solution:

Reimer Tiemann's reaction -

The electrophile used is : Ccl_2 (dichlorocarbone) -CHO is introduced in the ortho position.

- wherein



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The reaction of phenol with chloroform in the presence of dilute sodium hydroxide finally introduces aldehyde -CHO functional group and hydroxy group at -ortho position. The reaction is known as Reimer - Tiemann reaction.

Hence, the option number (4) is correct.

Q. 2 If the equilibrium constant for

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $1/2N_2(g) + 1/2O_2(g) \rightleftharpoons NO(g)$ will be

Option 1:
$K^{1/2}$

Option 2:
$\frac{K}{2}$

Option 3:
K

Option 4:
K^2

Correct Answer:
$K^{1/2}$

Solution:

$N_2 + O_2 \rightleftharpoons 2NO, K$

$\frac{1}{2}N_2 + \frac{1}{2}O_2 \rightleftharpoons NO, K' = \sqrt{K}$

Or

If the reaction is multiplied by 1/2, then the new equilibrium constant, $K' = K^{1/2}$

Hence, the answer is the option (1).

Q. 3 The number of water molecules is maximum in :

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Option 1:
18 molecules of water

Option 2:
1.8 gram of water

Option 3:
18 gram of water

Option 4:
18 moles of water

Correct Answer:
18 moles of water

Solution:
Mole Concept -

one mole = 6.0221367×10^{23} = N_A

Let us check the number of molecules in each option.

18 molecules of water have 18 molecules

1.8 grams of water has $\frac{1.8}{18} \times \mathrm{N_A} = 0.1\ \mathrm{N_A}$ molecules

18 grams of water has $\frac{18}{18}\ \mathrm{N_A} = \mathrm{N_A}$ molecules

18 moles of water has 18 N_A molecules

Hence, the answer is the option (4).

Q. 4 The formation of the oxide ion O^{2-}(g), from oxygen atom requires first an exothermic and then an endothermic step as shown below:

$$O(g) + e^- \rightarrow O^-(g); \Delta_f H^\ominus = -141kJ\ mol^{-1}$$

$$O^-(g) + e^- \rightarrow O^{2-}(g); \Delta_f H^\ominus = +780kJ\ mol^{-1}$$

Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that

Option 1:
Electron repulsion outweighs the stability gained by achieving noble gas configuration.

Option 2:
O^- ion has comparatively smaller size than oxygen atom.

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Option 3:
Oxygen is more electronegative.

Option 4:
Addition of electron in oxygen results in larger size of the ion.

Correct Answer:
Electron repulsion outweighs the stability gained by achieving noble gas configuration.

Solution:
The first electron gain enthalpy of the Oxygen atom is negative, as when the electron is added, the attractive forces between the incoming electron and nucleus outweigh the repulsions in between electrons.

But on adding the second electron, the repulsions take over attractive forces, hence it's an endothermic process.

Hence, the answer is the option (1).

Q. 5 What is the mole fraction of the solute in a 1.00 m aqueous solution ?

Option 1:
0.177

Option 2:
1.770

Option 3:
0.0354

Option 4:
0.0177

Correct Answer:
0.0177

Solution:
As we learned in

$$\text{Molality} = \frac{\text{Moles of solute}}{\text{Mass of solution(kg)}}$$

$$\text{No. moles of water in 1000g} = \frac{1000}{8} = 55.5$$

$$\text{No. of moles of solute} = 1$$

$$\text{Mole fraction of solute} = \frac{1}{1+55.5} = 0.0177$$

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Hence, the answer is the option (4).

Q. 6 The rate constant of the reaction $A \rightarrow B$ is $0.6 \times 10^{-3} mole\ per\ second$. If the concentration of A is 5 M, then the concentration of B after 20 minutes is:

Option 1:
1.08 M

Option 2:
3.60 M

Option 3:
0.36 M

Option 4:
0.72 M

Correct Answer:
0.72 M

Solution:
For a zero-order reaction,

$$A_t = A_o + kt$$

A_o is the initial concentration,

A

The reaction is of zero order as the unit of rate constant is mol L^{-1} s^{-1}

$\therefore$ concentration = K x t

= 0.6 x 10^{-3} x 20 x 60 M

= 0.72 M

Hence, the answer is the option (4).

Q. 7 Decreasing order of stability of O_2 ,O_2^-, O_2^+ and O_2^{2-} is :

Option 1:
$O_2^+ > O_2 > O_2^- > O_2^{2-}$

Option 2:
$O_2^{2-} > O_2^- > O_2 > O_2^+$

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Option 3:

$O_2 > O_2^+ > O_2^{2-} > O_2^-$

Option 4:

$O_2^- > O_2^{2-} > O_2^+ > O_2$

Correct Answer:

$O_2^+ > O_2 > O_2^- > O_2^{2-}$

Solution:

Bond Order -

Bond order is defined as one-half the difference between the number of electrons present in the bonding and the antibonding orbitals.

- wherein

$$Bond\,Order = \frac{N_b - N_a}{2}$$

$$Bonder\ order = \frac{Nb - Na}{2}$$

$$For\ O_2^+ \ :\ Bond\ order\ = \frac{10-5}{2} = 2.5$$

$$For\ O_2^- \ :\ Bond\ order\ = \frac{10-7}{2} = 1.5$$

$$For\ O_2^{2-} \ :\ Bond\ order\ = \frac{10-8}{2} = 1.0$$

$$For\ O_2 \ :\ Bond\ order\ = \frac{10-6}{2} = 2$$

$$Since\ bond\ order\ of\ O_2^+ > B.O.\ of\ O_2 > B.O.\ of\ O_2^- > B.O\ of\ O_2^{2-}$$

Hence order of stability

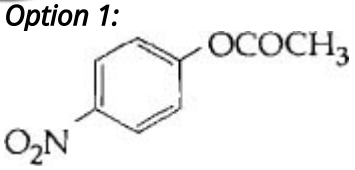
$$O_2^+ > O_2 > O_2^- > O_2^{2-}$$

since Bond order is directly proportional to stability.

Hence, the answer is the option (1).

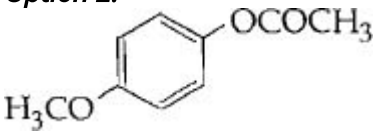
Q. 8 Which one of the following esters gets hydrolysed *most easily* under alkaline conditions ?

Option 1:

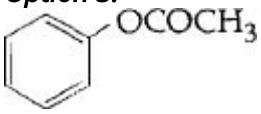


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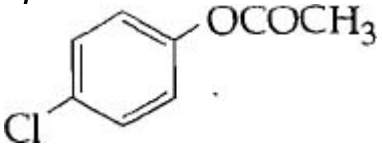
Option 2:



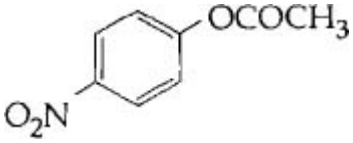
Option 3:



Option 4:



Correct Answer:

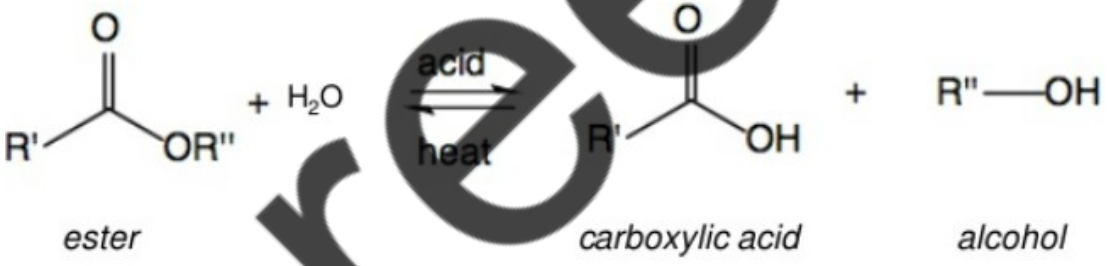


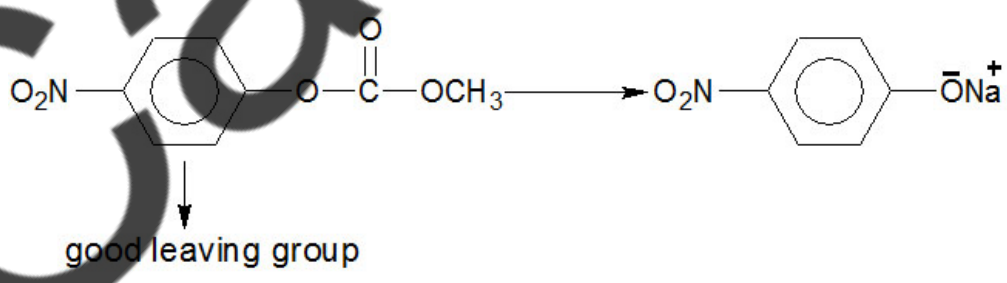
Solution:

Acidic and basic hydrolysis of Esters -

Acidic hydrolysis gives carboxylic acid directly. Basic hydrolysis gives carboxylates which acidify to give the carboxylic acid.

- wherein





Q. 9 On heating which of the following releases CO_2 most easily ?

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Option 1:
K_2CO_3

Option 2:
Na_2CO_3

Option 3:
$MgCO_3$

Option 4:
$CaCO_3$

Correct Answer:
$MgCO_3$

Q. 10 Which one of the following pairs of solution is not an acidic buffer ?

Option 1:
$HClO_4$ and $NaClO_4$

Option 2:
CH_3COOH and CH_3COONa

Option 3:
H_2CO_3 and Na_2CO_3

Option 4:
H_3PO_4 and Na_3PO_4

Correct Answer:
$HClO_4$ and $NaClO_4$

Solution:
$HClO_4$ Completely dissociates in an aqueous solution because it is a strong acid. Its solution with its salt cannot form a buffer as it can't resist changes in pH.

An acidic buffer is a mixture of a weak acid and its salt with a strong base. HClO4 is a strong acid.

Hence, the answer is the option (1).

Q. 11 The sum of coordination number and oxidation number of the metal M in the complex $[M(en)_2 (C_2O_4)]Cl$ (where *en* is ethylenediamine) is:

Option 1:
9

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Option 2:
6

Option 3:
7

Option 4:
8

Correct Answer:
9

Solution:
$[M(en)_2\,(C_2O_4)]Cl$

Coordination Number =6
Oxidation Number of M=+3

Hence, the answer is the option (1).

Q. 12 Which of the statements given below is **incorrect**?

Option 1:
Cl_2O_7 is an anhydride of perchloric acid.

Option 2:
O_3 molecule is bent

Option 3:
ONF is isoelectronic with $O2N^-$

Option 4:
OF_2 is an oxide of fluorine

Correct Answer:
OF_2 is an oxide of fluorine

Solution:
Normal oxides - These contain oxygen atoms according to the normal oxidation number

MgO, H_2O, CaO, Li_2O

Cl_2O_7 is an anhydride of $HClO_4$

O_3 molecule is bent shaped.

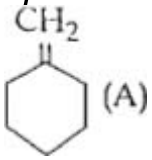
OF_2 is fluoride of oxygen.

Hence, the correct answer is option 4.

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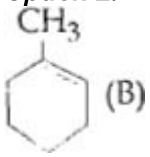
Q. 13 In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is:

Option 1:



(A)

Option 2:



(B)

Option 3:

(A) and (B)

Option 4:



Correct Answer:

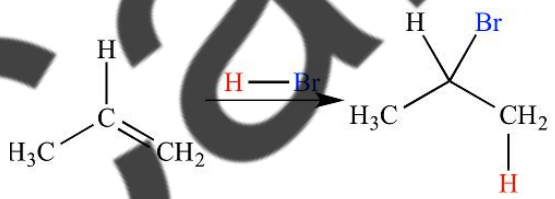
(A) and (B)

Solution:

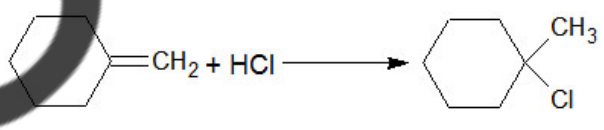
Markovnikov's Rule -

Negative part of the reagent attacks on that carbon of multiple bond which have less number of hydrogen atoms.

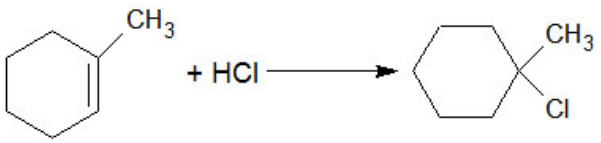
- wherein



(A)



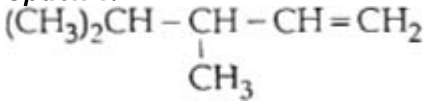
(B)



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Q. 14 2, 3 - Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid ?

Option 1:



Option 2:

$(CH_3)_3\,C - CH = CH_2$

Option 3:

$(CH_3)_2\,C = CH - CH_2 - CH_3$

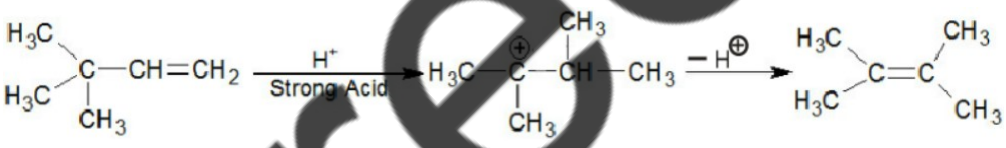
Option 4:

$(CH_3)_2\,CH - CH_2 - CH = CH_2$

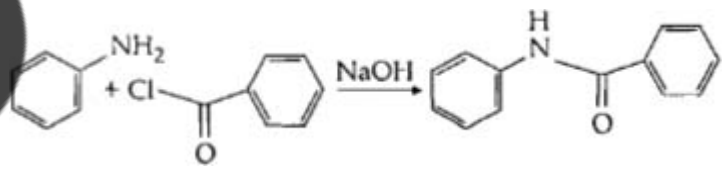
Correct Answer:

$(CH_3)_3\,C - CH = CH_2$

Solution:



Q. 15 The following reaction



is known by the name:

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Option 1:
Friedel-Craft's reaction

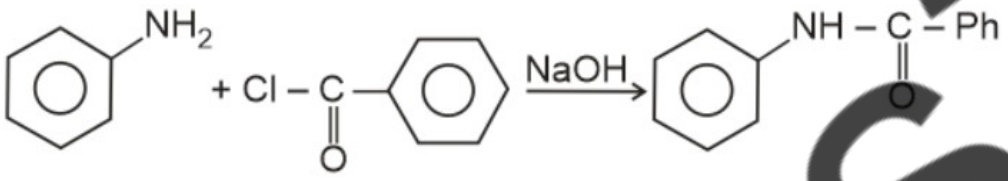
Option 2:
Perkin's reaction

Option 3:
Acetylation reaction

Option 4:
Schotten-Baumann reaction

Correct Answer:
Schotten-Baumann reaction

Solution:



Reaction is known as
Schotten – Baumann reaction

Q. 16 In the extraction of copper from its sulphide ore, the metal is finally obtained by the reduction of cuprous oxide with:

Option 1:
iron (II) sulphide

Option 2:
carbon monoxide

Option 3:
copper (I) sulphide

Option 4:
sulphur dioxide

Correct Answer:
copper (I) sulphide

Solution:
Reduction of Metal Oxide to Metal - The crude metal is obtained when the roasted ore or calcined ore is put to reduction by using a suitable reducing agent. In the case of copper, Cu_2S, it undergoes self reduction Bessemer converter with Cu_2O.to get blister copper.

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$$\underset{Copper(I)Sulphide}{Cu_2S} + \underset{Copper\ oxide}{2Cu_2O} \longrightarrow \underset{Copper}{6Cu} + SO_2$$

Hence, the option number (3) is correct.

Q. 17 If Avogadro number $N_{A,}$ is changed from 6.022 x 10^{23} mol^{-1} to 6.022 x 10^{20} mol^{-1}, this would change :

Option 1:
the definition of mass in units of grams.

Option 2:
the mass of one mole of carbon.

Option 3:
the ratio of chemical species to each other in a balanced equation.

Option 4:
the ratio of elements to each other in a compund.

Correct Answer:
the mass of one mole of carbon.

Solution:
As we learnt in

One mole is the amount of a substance that contains as many particles or entities as there are atoms in exactly 12 g (or 0.012 kg) of the 12C isotope.

1 mole of substance = 6.0221367 X 10^{23} **units** of that substance (such as atoms, molecules, or ions). The number 6.0221367 X 10^{23} is known as Avogadro's number or Avogadro's constant.

1 mol of carbon has mass = 12 g

6.022×10^{23} atoms of carbon mass = 12 g

So if now, 1 mole = 6.022 $\times 10^{20}$

then 6.022 $\times 10^{20}$ atoms of carbon mass = $\frac{12g}{1000}$

$\Rightarrow$ 1 mol of carbon atom mass = 0.012 g

Hence, the answer is the option (2).

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Q. 18 The variation of the boiling points of the hydrogen halides is in the order $HF > HI > HBr > HCl.$

What explains the higher boiling point of hydrogen fluoride?

Option 1:
The electronegativity of fluorine is much higher than for other elements in the group.

Option 2:
There is strong hydrogen bonding between HF molecules.

Option 3:
The bond energy of HF molecules is greater than in other hydrogen halides.

Option 4:
The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule.

Correct Answer:
There is strong hydrogen bonding between HF molecules.

Solution:
The boiling point of Hydride -

$$HF > HI > HBr > HCl$$

Going down the boiling point must increase with the increase in molar mass of halogen but HF has a higher boiling point than that of HI, this is due to very strong H- bonding present in HF.
The electronegativity difference between H and F is very large which makes strong H- bonds

The Correct answer is option 2.

Q. 19 Which of the following reaction(s) can be used for the preparation of alkyl halides?

(I) $CH_3CH_2OH + HCl \overset{anh.ZnCl_2}{\rightarrow}$

(II) $CH_3CH_2OH + HCl \rightarrow$

(III) $(CH_3)_3COH + HCl \rightarrow$

(IV) $(CH_3)_2CHOH + HCl \overset{anh.ZnCl_2}{\rightarrow}$

Option 1:
(I), (III) and (IV) only

Option 2:
(I) and (II) only

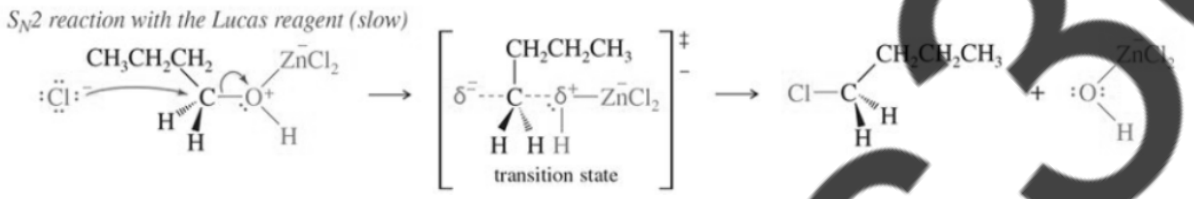
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Option 3:
(IV) only

Option 4:
(III) and (IV) only

Correct Answer:
(I), (III) and (IV) only

Solution:
Lucas's reagent contains zinc chloride and concentrated hydrochloric acid. This reagent reacts with primary, secondary and tertiary alcohols at different rates. Tertiary alcohols react almost instantaneously, secondary alcohols react in about 1-5 minutes and primary alcohols react very slowly. The reaction may take 10 minutes to several days



HCl reacts with 1° and 2° alcohols to yield the corresponding chloride only in the presence of anhydrous $ZnCl_2$, Thus (I) and (IV) can be used to prepare alkyl halide due to the presence of anhydrous $ZnCl_2$, while (III) gives alkyl halide due to formation of more stable carbonation (3° Carbocation).

(II) with HCl and anhydrous $ZnCl_2$,reaction very slow. So (II) will not react due to absence of $ZnCl_2$,

(1) $CH_3—CH_2—OH + HCl \xrightarrow[\text{Lucas reagent}]{\text{an hyd } ZnCl_2} CH_3—CH_2—Cl + H_2O$

(2) $CH_3—CH_2—OH + HCl \xrightarrow[\text{Temp}]{\text{Room}} \text{No reaction}$

(3) $H_3C—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—OH \xrightarrow[\text{R.T.}]{HCl} H_3C—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—Cl + H_2O$

(4) $H_3C—\underset{\displaystyle CH_3}{\overset{\displaystyle H}{\overset{|}{\underset{|}{C}}}}—OH \xrightarrow[ZnCl_2]{HCl} H_3C—\underset{\displaystyle CH_3}{\overset{\displaystyle H}{\overset{|}{\underset{|}{C}}}}—Cl + H_2O$

Hence, option number (1) is correct.

Q. 20 The name of complex ion, $[Fe(CN)_6]^{3-}$ is:

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Option 1:
Hexacyanoiron (III) ion

Option 2:
Hexacyanitoferrate (III) ion

Option 3:
Tricyanoferrate (III) ion

Option 4:
Hexacyanidoferrate (III) ion

Correct Answer:
Hexacyanidoferrate (III) ion

Solution:
The name of $[Fe(CN)_6]^{3-}$ is Hexacyanidoferrate (III) ion.

Hence, the answer is the option (4).

Q. 21 Assuming complete ionization, the same moles of which of the following compounds will require the *least* amount of acidified $KMnO_4$ for complete oxidation?

Option 1:
$FeSO_4$

Option 2:
$FeSO_3$

Option 3:
FeC_2O_4

Option 4:
$Fe(NO_2)_2$

Correct Answer:
$FeSO_4$

Solution:
In $FeSO_4$, $KMnO_4$ only required for oxidation of Fe^{2+} because in SO_4^{2-} sulphur is in its maximum oxidation state so $FeSO_4$ requires the least number of moles of $KMnO_4$ for oxidation.

In the case of $FeSO_3$, sulfur has an oxidation state of +4 and will oxidize to SO_4^{2-} ion having an oxidation state of +6.

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For, FeC_2O_4, C will oxidize to CO_2 from oxalate ion changing oxidation state from +3 to +4.

Similarly, $Fe(NO_2)_2$, Nitrite ions will oxidize to nitrate ions changing the oxidation state of N from +3 to +5.

Hence, the answer is the option (1).

Q. 22 In which of the following pairs, both the species are not isostructural ?

Option 1:
$SiCl_4$, PCl_4^+

Option 2:
diamond, silicon carbide

Option 3:
NH_3, PH_3

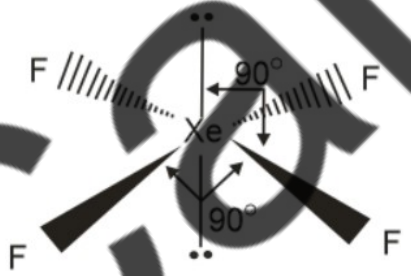
Option 4:
Xe F_4, XeO_4

Correct Answer:
Xe F_4, XeO_4

Solution:
Structure of Xenon tetrafluoride -

Square planar and hybridisation is Sp^3d^2

- wherein



XeF_4 has square planar structure with hybridisation sp^3d^2 while XeO_4 has tetrahedral structure with hybrdisation sp^3.

Correct option is 4.

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Q. 23 Caprolactam is used for the manufacture of:

Option 1:
Nylon - 6

Option 2:
Teflon

Option 3:
Terylene

Option 4:
Nylon - 6, 6

Correct Answer:
Nylon - 6

Solution:
Nylon 6 -

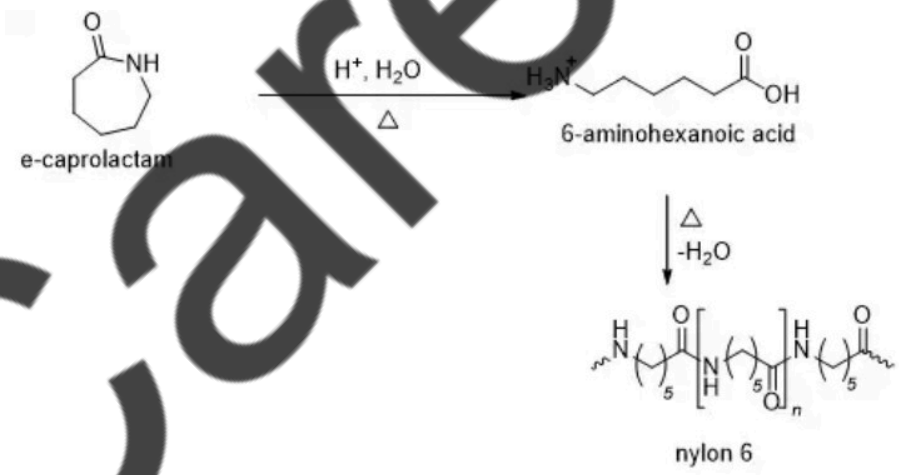
- Obtained by heating caprolactam with water.

- Homopolymer

- wherein

- High tensile strength

- Used in manufacture of tyre, cords, fabrics and ropes.



Caprolactam is the monomer of Nylon-6.

Hence, the option number (1) is correct.

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Q. 24 The hybridization involved in complex $[Ni(CN)_4]^{2-}$ is: (At. No.Ni = 28)

Option 1:
dsp^2

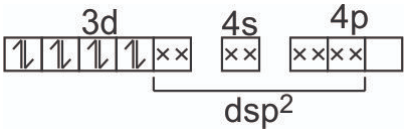
Option 2:
sp^3

Option 3:
d^2sp^2

Option 4:
d^2sp^3

Correct Answer:
dsp^2

Solution:
In $[Ni\ (CN)_4]^{2-}$ The oxidation state of Ni is +2 and CN^- is a strong field ligand which can pair the unpaired electrons of Ni^{2+}



Q. 25 What is the mass of the precipitate formed when 50 mL of 16.9% solution of $AgNO_3$ is mixed with 50 mL of 5.8% NaCl solution ?(Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)

Option 1:
28 **g**

Option 2:
3.5 **g**

Option 3:
7 g

Option 4:
14 **g**

Correct Answer:
7 g

Solution:

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As we learnt in

Mass percent of solution is mass of solute present in 100 gram solution.

Mass percent = ((mass of solute)/(mass of solution)) X 100

Now,

16.9% solution of $AgNO_3$ means 16.9g $AgNO_3$ in 100ml solution.

$\Rightarrow$ 50ml of solution contains $\frac{16.9}{2} = 8.45g$

Similarly, 100ml of other solution contains 5.8g NaCl

50ml of other solution contains 2.9g NaCl

Mol of $AgNO_3$ = $\frac{8.45}{169.8} = 0.049$

Mol of NaCl = $\frac{2.9}{58.5} = 0.049$

$AgNO_3$	+	NaCl	$\rightarrow$	AgCl	+	$NaNO_3$
0.049		0.049		0		0
0		0		0.049		0.049

Mass of AgCl = 0.049 $\times$ Molar mass of AgCl

= 0.049 $\times$ 143.5 = 7g

Therefore, the correct option is (3).

Q. 26 Gadolinium belongs to the 4f series. Its atomic number is 64. Which of the following is the **correct** electronic configuration of gadolinium?

Option 1:
$[Xe]4f^86s^2$

Option 2:
$[Xe]4f^96s^1$

Option 3:
$[Xe]4f^75d^16s^2$

Option 4:
$[Xe]4f^65d^26s^2$

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Correct Answer:

$[Xe]4f^{7}5d^{1}6s^{2}$

Solution:

Electronic configuration -

The lanthanoid follows the $4f^{1-14}5d^{0-1}6s^{2}$ configuration (the common configuration)with some exceptions due to the full-filled and half-filled electronic configuration.

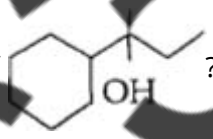
$eg \rightarrow 4f^{1}5d^{1}6s^{2}$

The electronic configuration of Gadolinium is

$[Xe]4f^{7}5d^{1}6s^{2}$

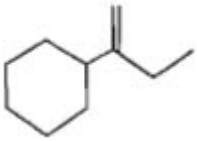
Hence, the answer is the option (3).

Q. 27

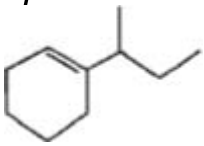
Which of the following is not the product of dehydration of 

?

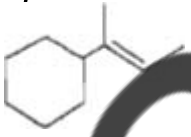
Option 1:



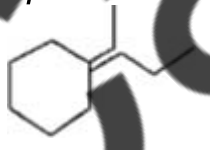
Option 2:



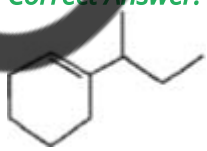
Option 3:



Option 4:



Correct Answer:



Solution:

As we learnt in

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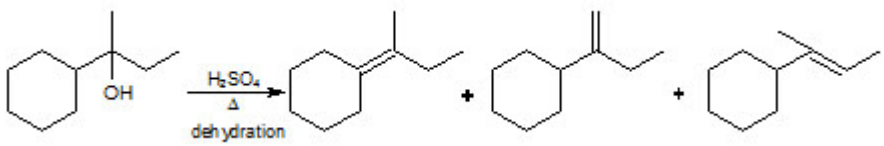
Dehydration of alcohols -

Ease of dehydration

$3^\circ > 2^\circ > 1^\circ$

- wherein

$$CH_3CH_2OH + Conc.H_2SO_4 \overset{443K}{\rightarrow} CH_2 = CH_2 + H_2O$$



Q. 28 A gas such as carbon monoxide would be most likely to obey the ideal gas law at:

Option 1:
High temperatures and low pressure.

Option 2:
Low temperatures and high pressure.

Option 3:
High temperatures and high pressure.

Option 4:
Low temperatures and low pressure.

Correct Answer:
High temperatures and low pressure.

Solution:
An ideal gas is a gas that has no interaction between the constituent particles and follows gas laws.

A real gas can behave as an ideal gas if the volume of a molecule is negligible and interaction between the molecule is negligible. This will happen when the number of molecules per unit volume is very small.

From the ideal gas equation,

$$PV = nRT$$

$$or\ \frac{V}{n} = \frac{RT}{P}$$

$\frac{V}{n}$ will be very large if T is very large and P is very small.

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Hence, option 1 is correct.

Q. 29 The stability of the $+1$ oxidation state among $Al, Ga, In \text{ and } Tl$ increases in the sequence :

Option 1:
$Ga < In < Al < Tl$

Option 2:
$Al < Ga < In < Tl$

Option 3:
$Tl < In < Ga < Al$

Option 4:
$In < Tl < Ga < Al$

Correct Answer:
$Al < Ga < In < Tl$

Solution:
Inert pair effect - The phenomenon in which the outer shell s electrons (ns^2) penetrate to $(n-1)d$-electrons and thus become closer to the nucleus and are more effectively pulled by the nucleus.

This results in less availability of ns^2 electrons pair for bonding or ns^2 electron pair becomes inert.

On moving down the group stability of the +1 oxidation state increases in group 13 elements.

order of stability of +1 oxidation state $B < Al < Ga < In < Tl$

Hence, the answer is the option (2).

Q. 30 What is the pH of the resulting solution when equal volumes of $0.1MNaOH \text{ and } 0.01MHCl$ are mixed ?

Option 1:
12.65

Option 2:
2.0

Option 3:
7.0

Option 4:
1.04

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Correct Answer:

12.65

Solution:

Let volumes of each solution be 1L

moles of NaOH=0.1mol

moles of HCl=0.01mol

moles of NaOH left after mixing =0.09ml

Total volume = 2L

Molarity = Normality (as n=1)=$\frac{0.09}{2} = 0.045M \ or \ 0.045N$

$pOH = -log(0.045) = 1.35$

$pH = 14 - pOH = 14 - 1.35 = 12.65$

Hence, the answer is the option (1).

Q. 31 Strong reducing behaviour of H_3PO_2 is due to :

Option 1:

Presence of one - OH group and two P - H bonds

Option 2:

High electron gain enthalpy of phosphorus

Option 3:

High oxidation state of phosphorus

Option 4:

Presence of two - OH groups and one P - H bond

Correct Answer:

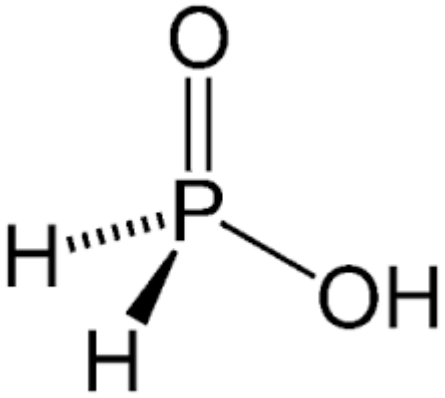
Presence of one - OH group and two P - H bonds

Solution:

Hypophosphorous Acid -

H_3PO_2 (+1), monobasic

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The strong reducing behavior of H_3PO_2 is due to the presence of one -OH group and two P-H bonds.

The more the number of P-H bonds, the lower is the oxidation state of P, and the more is reducing power.

The correct answer is option 1

Q. 32 The number of structural isomers possible from the molecular formula C_3H_9N is :

Option 1:
4

Option 2:
5

Option 3:
2

Option 4:
3

Correct Answer:
4

Solution:

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The possible structural isomers are :

1. $CH_3CH_2CH_2NH_2$
   Propanamine
2. $CH_3-\underset{\displaystyle NH_2}{\underset{|}{CH}}-CH_3$
   Isopropylamine
3. $CH_3CH_2NHCH_3$
   N-Methylethanamine
4. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{N}}-CH_3$
   N,*N*-Dimethylmethanamine

Hence, the option number (1) is correct.

Q. 33 Which of the following statements is **not** correct for a nucleophile ?

Option 1:
Nucleophile is a Lewis acid

Option 2:
Ammonia is a nucleophile

Option 3:
Nucleophiles attack low e^- density sites

Option 4:
Nucleophiles are not electron seeking

Correct Answer:
Nucleophile is a Lewis acid

Solution:
Nucleophile -

Those e-rich chemical species have the capacity to attack the e-deficient portion of the substrate.

- wherein

Nucleophiles must have complete octet and at least one lone pair

Nucleophiles are those having the capacity to donate electrons so they are Lewis bases.

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Hence, the answer is the option (1).

Q. 34 Number of possible isomers for the complex $[Co(en)_2Cl_2]$ Cl will be: (en = ethylenediamine)

Option 1:
2

Option 2:
1

Option 3:
3

Option 4:
4

Correct Answer:
3

Solution:
The complex $[Co(en)_2Cl_2]Cl$ will possess two geometrical isomers (cis and trans) and its cis isomer is optically active that's why the total number of isomers in $[Co(en)_2Cl_2]Cl$ is three.

Hence, the answer is the option (3).

Q. 35 Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?

(At. no. Z = 22)

Option 1:
3s 4s 3p 3d

Option 2:
4s 3s 3p 3d

Option 3:
3s 3p 3d 4s

Option 4:
3s 3p 4s 3d

Correct Answer:
3s 3p 4s 3d

Solution:
Electronic configuration of d block elements -

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These elements have other electronic configurations of $(n-1)d^{1-10}ns^{1-2}$

$Ti(22) = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

correct order is $3s\ 3p\ 4s\ 3d$

Hence, the answer is the option (4).

Q. 36 In an S_N1 reaction on chiral centres, there is :

Option 1:
100% racemization

Option 2:
inversion more than retention leading to partial racemization

Option 3:
100% retention

Option 4:
100% inversion

Correct Answer:
inversion more than retention leading to partial racemization

Solution:
In SN^1, inversion more than retention leads to partial recemization. due to intimate-ion pair.

Hence, the answer is the option (2).

Q. 37 The vacant space in bcc lattice unit cell is :

Option 1:
26%

Option 2:
48%

Option 3:
23%

Option 4:
32%

Correct Answer:
32%

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Solution:
As we learned in

In a BCC lattice, the total number of atoms is 2.

So,

$$\text{Packing efficiency } = \frac{\text{Volume occupied by two spheres in the unit cell } \times 100}{\text{Total volume of the unit cell}}\%$$

$$\text{Packing efficiency } = \frac{2 \times (4/3)\pi r^3 \times 100}{[(4/\sqrt{3})r]^3}\%$$

$$\text{Packing efficiency } = \frac{(8/3)\pi r^3 \times 100}{64/(3\sqrt{3})r^3}\% = 68\%$$

Now, 68% of all volume is occupied by spheres.

So, the vacant space = 100 - 68 = 32%.

Therefore, the correct option is (4).

Q. 38 The heat of combustion of carbon to CO_2 is -393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is :

Option 1:
- 315 kJ

Option 2:
+ 315 kJ

Option 3:
- 630 kJ

Option 4:
- 3.15 kJ

Correct Answer:
- 315 kJ

Solution:
Given,

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)};\ \Delta H_f = -393.5\ kJmol^{-1}$$

$\therefore$ the heat released on formation of 44g or 1 mole of CO_2 = -393.5 $kJmol^{-1}$

So heat released by the formation of 35.2g of carbon dioxide is

$$= \frac{-393.5 \times 35.2}{44} = -315\ kJmol^{-1}$$

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Hence, the answer is the option (1).

Q. 39 Aqueous solution of which of the following compounds is the best conductor of electric current ?

Option 1:
Acetic acid, $C_2H_4O_2$

Option 2:
Hydrochloric acid, HCl

Option 3:
Ammonia, NH_3

Option 4:
Fructose, $C_6H_{12}O_6$

Correct Answer:
Hydrochloric acid, HCl

Solution:
In an Aqueous solution, HCl is the best conductor of electric current. Since HCl is a strong acid and hence it ionizes easily in an aqueous solution and provides free ions which are the carrier of electricity.

Hence, the answer is the option (2).

Q. 40 The oxidation of benzene by V_2O_5 in the presence of air produces:

Option 1:
benzoic anhydride

Option 2:
maleic anhydride

Option 3:
benzoic acid

Option 4:
benzaldehyde

Correct Answer:
maleic anhydride

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Q. 41 Reaction of a carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is :

Option 1:
a Grignard reagent

Option 2:
hydrazine in presence of feebly acidic solution

Option 3:
hydrocyanic acid

Option 4:
sodium hydrogen sulphite

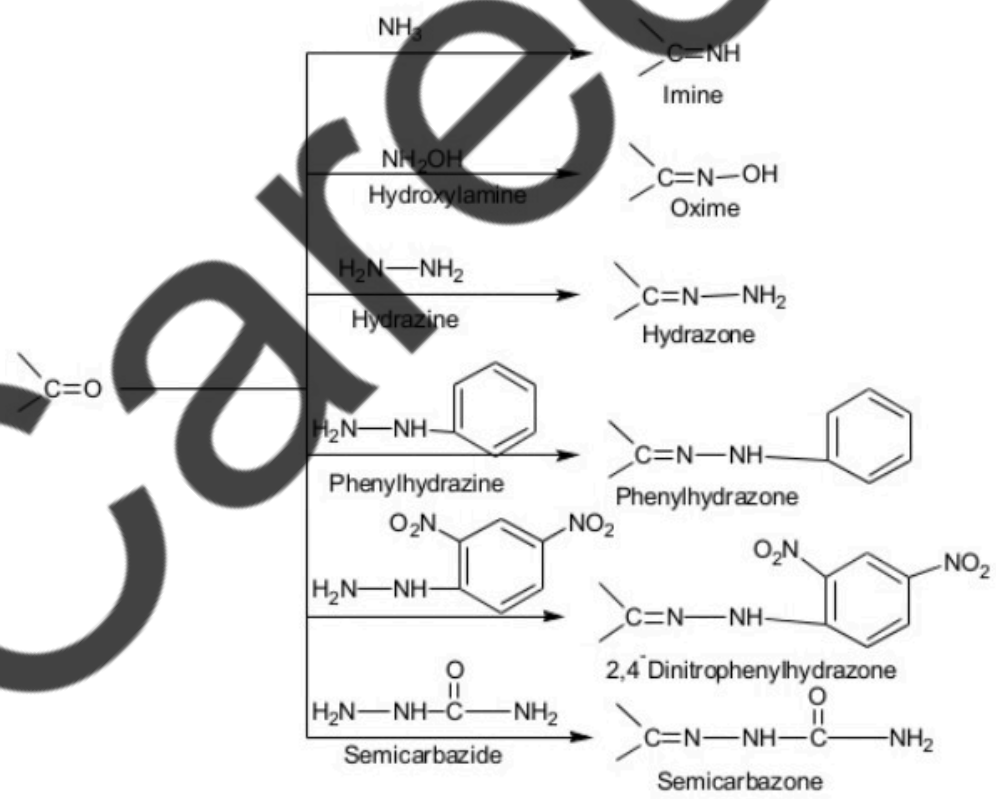
Correct Answer:
hydrazine in presence of feebly acidic solution

Solution:
Addition of ammonia and its derivatives with aldehydes and ketones -

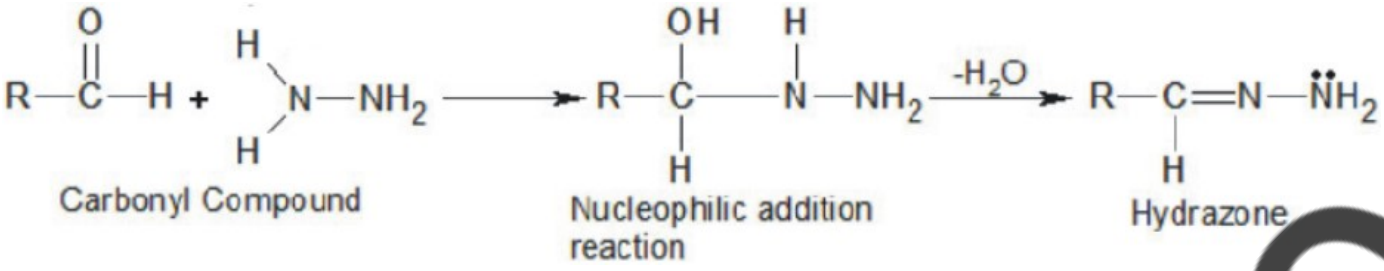
Product formation is favoured due to rapid dehydration of the intermediate to form > C = NG

$$G \ may \ be \ -OH, \ -NH_2, \ -NHC_6H_5, \ -NHCONH_2, \ etc$$

- wherein



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Q. 42 Method by which Aniline cannot be prepared is :

Option 1:
hydrolysis of phenylisocyanide with acidic solution.

Option 2:
degradation of benzamide with bromine in alkaline solution.on.

Option 3:
reduction of nitrobenzene with H_2/Pd in ethanol.

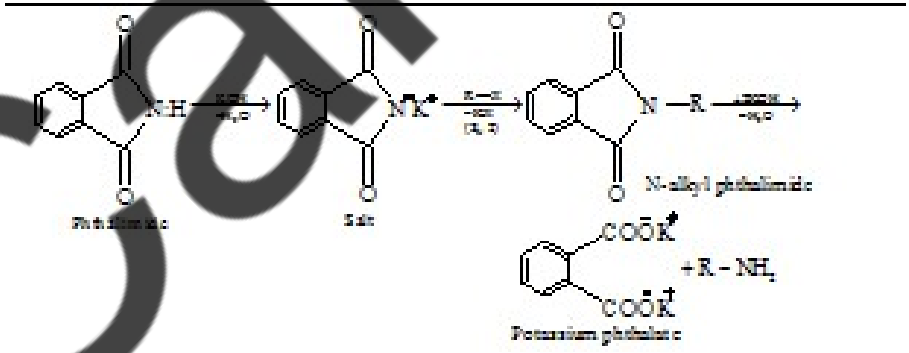
Option 4:
potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution.

Correct Answer:
potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution.

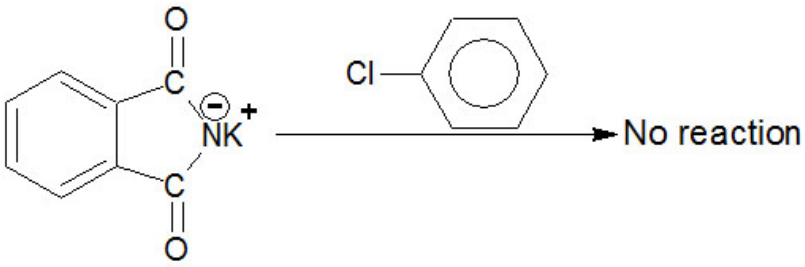
Solution:
Gabriel Phthalimide Synthesis -

Further alkylation can be stopped and a pure primary amine can be obtained by alkylation of phthalimide (Gabriel synthesis) followed by hydrolysis.

- wherein



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Because there exist partial double between chlorine and carbon.

Q. 43 Two possible stereo-structures of $CH_3CHOH.COOH$, which are optically active, are called:

Option 1:
Diastereomers

Option 2:
Atropisomers

Option 3:
Enantiomers

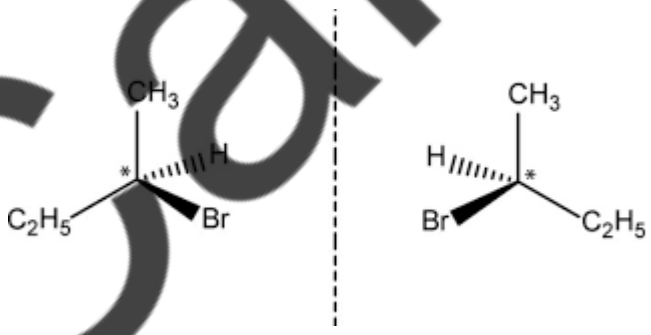
Option 4:
Mesomers

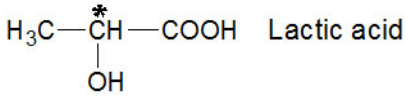
Correct Answer:
Enantiomers

Solution:
Optically active compound -

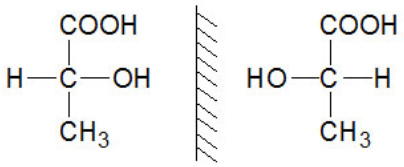
Those compounds having capacity to rotate plane polarised light is called optically active compound.

- wherein





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These are mirror image non-super imposible to each other are called enantiomers

Q. 44 The correct statement regarding defects in crystalline solids is :

Option 1:
Schottky defects have no effect on the density of crystalline solids.

Option 2:
Frenkel defects decrease the density of crystalline solids.

Option 3:
Frenkel defect is a dislocation defect.

Option 4:
Frenkel defect is found in halides of alkaline.

Correct Answer:
Frenkel defect is a dislocation defect.

Solution:
Schottky defect decreases the density of crystalline solids.

Frenkel defect does not change the density of the solid.

Frenkel defect is also called dislocation defect.

Frenkel defect is shown by ionic substances in which there is a large difference in the size of ions. for example, ZnS, AgCl, AgBr and AgI. So, the Frenkel defect does not found in halides of alkaline.

So, Only Statement II is correct.

Therefore, the correct option is (2).

Biology

Q. 1 In his classic experiments on pea plants, Mendel did not use :

Option 1:
Pod length

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Option 2:
Seed shape

Option 3:
Flower position

Option 4:
Seed colour

Correct Answer:
Pod length

Solution:
As we learnt in

Mendel's experiment -

Mendel investigated characters in the garden pea plants thet were manifested as two opposing traits eg- dwarf plants, yellow or green seeds.

- wherein

This allowed him to set up a basic frame-work of rules governing inheritance.

Inheritance of two genes -

Mendel crossed plants that differed in two characters colour and shape.

- wherein

As a result, yellow color was dominant over green and round shape dominant over wrinkled.

Mendel investigated characters in the garden pea plants thet were manifested as two opposing traits eg- dwarf plants, yellow or green seeds. He also crossed plants that differed in two characters colour and shape.

Q. 2 Which one of the following is not applicable to RNA ?

Option 1:
5' phosphoryl and 3' hydroxyl ends

Option 2:
Heterocyclic nitrogenous bases

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Option 3:
Chargaff 's rule

Option 4:
Complementary base pairing

Correct Answer:
Chargaff 's rule

Solution:
Chargaff's rules state that DNA from any cell of all organisms should have a 1:1 ratio (base Pair Rule) of pyrimidine and purine bases and, more specifically, that the amount of guanine should be equaled to cytosine and the amount of adenine should be equaled to thymine.

Q. 3 Male gametophyte in angiosperms produces :

Option 1:
Single sperm and a vegetative cell

Option 2:
Single sperm and two vegetative cells

Option 3:
Three sperms

Option 4:
Two sperms and a vegetative cell

Correct Answer:
Two sperms and a vegetative cell

Solution:
As we discussed in

Pollen grains -

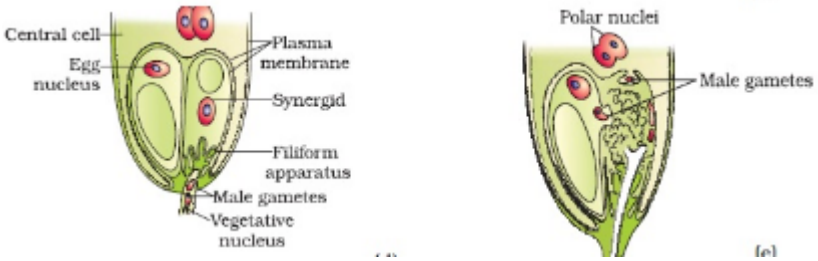
The pollen grains represents the male gametophytes.

-pollen-pistil interaction - Events occuring post Compatible pollination -

In plants, where the pollen grains are shed at two-celled condition (a vegetative cell and a generative cell), the generative cell divides and forms the two male gametes during the growth of pollen tube in the stigma. In plants which shed pollen in the three-celled condition, pollen tubes carry the two male gametes from the beginning. Pollen tube, after reaching the ovary, enters the ovule through the micropyle and then enters one of the synergids through the filiform apparatus

- wherein

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Pollen grain are male gametophyte having a vegetative cell and a generative cell (forms two males gametes, i.e. sperms)

Q. 4 Which of the following are not membrane - bound ?

Option 1:
Ribosomes

Option 2:
Lysosomes

Option 3:
Mesosomes

Option 4:
Vacuoles

Correct Answer:
Ribosomes

Solution:
As we learnt in

Ribosomes -

It is a complex made up of RNA and protein and is the site for protein synthesis.

- wherein

Several ribosomes may attach to a single mRNA and form chains called polyribosomes or polysome.

Ribosomes are complex of RNA and protein.

Q. 5 The chitinous exoskeleton of arthropods is formed by the polymerisation of :

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Option 1:
D - glucosamine

Option 2:
N - acetyl glucosamine

Option 3:
lipoglycans

Option 4:
keratin sulphate and chondroitin sulphate

Correct Answer:
N - acetyl glucosamine

Solution:
As we learnt in

Exoskeleton of cockroach -

The body of cockroach is covered by thick hard chitinous cuticle, which is secreted by epidermis, forming the exoskeleton.

- wherein

Each segment of exoskeleton contains 4 layers called sclerites, the dorsal sclerites is called sternum or sternite; two lateral sclerites are called pleura or pleurites. The terga, sternum and pleura are joined by arthrodial membrane.

Exoskeleton of cockroach (arthropods) is composed of hard chitinous cuticle which is made up of N-acetyl glucosamine

---

Q. 6 Among china rose, mustard, brinjal, potato, guava, cucumber, onion and tulip, how many plants have superior ovary ?

Option 1:
Six

Option 2:
Three

Option 3:
Four

Option 4:
Five

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Correct Answer:
Six

Solution:
A flower has a superior ovary when the base of the ovary is located above where the sepals, petals, and stamens are attached. This point of attachment is referred to as the receptacle or hypanthium, the fused bases of the three floral parts. A superior ovary is found in - fleshy fruits such as true berries, drupes; members of family Cucurbitaceae like onion, melons and squash; brinjal, china rose, mustard, tomatoes, potato, oranges, tulips. Guava and cucumber have inferior ovary.

Q. 7 The function of the gap junction is to :

Option 1:
facilitate communication between adjoining cells by connecting the cytoplasm for rapid transfer of ions, small molecules and some large molecules.

Option 2:
separate two cells from each other.

Option 3:
stop substance from leaking across a tissue.

Option 4:
performing cementing to keep neighbouring cells together.

Correct Answer:
facilitate communication between adjoining cells by connecting the cytoplasm for rapid transfer of ions, small molecules and some large molecules.

Solution:
As we learnt in

Gap Junction -

Gap Junction - Connection of cytoplasma

-

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Q. 8 Which of the following immunoglobulins does constitute the largest percentage in human milk ?

Option 1:
Ig M

Option 2:
Ig A

Option 3:
Ig G

Option 4:
Ig D

Correct Answer:
Ig A

Solution:
As we learnt in

Colostrum -

The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant.

- wherein

The foetus also receives some antibodies from their mother, through the placenta during pregnancy. These are some examples of passive immunity.

All types of immunoglobulin are found in human milk. Out of these secretory IgA, a type of immunoglobuin that protects the ears, nose, throat and the gastrointestinal tract, is founded largest amount.

Q. 9 In mammalian eye, the 'fovea' is the center of the visual field, where:

Option 1:
the optic nerve leaves the eye.

Option 2:
only rods are present.

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Option 3:
more rods than cones are found.

Option 4:
high density of cones occur, but has no rods.

Correct Answer:
high density of cones occur, but has no rods.

Solution:
As we learnt in

Fovea -

At the posterior pole of the eye lateral to the blind spot, there is a yellowish pigmented spot called macula lutea with a central pit called the fovea.

- wherein

The fovea is a thinned-out portion of the retina where only the cones are densely packed. It is the point where the visual acuity (resolution) is the greatest.

Only cones are densely packed and rodes Hu are absent.

Q. 10 Doctors use stethoscope to hear the sounds produced during each cardiac cycle. The second sound is heard when :

Option 1:
Ventricular walls vibrate due to gushing in of blood from atria

Option 2:
Semilunar valves close down after the blood flows into vessels from ventricles

Option 3:
AV node receives signal from SA node

Option 4:
AV valves open up

Correct Answer:
Semilunar valves close down after the blood flows into vessels from ventricles

Solution:
As we learnt in

Lub-dub -

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During each cardiac cycle two prominent sounds are produced which can be easily heard through a stethoscope.

- wherein

The first heart sound (lub) is associated with the closure of the tricuspid and bicuspid valves whereas the second heart sound (dub) is associated with the closure of the semilunar valves.

The second heart sound (dub) is associated with the closure of the semilunar value.

Q. 11 Coconut water from a tender coconut is:

Option 1:
Free nuclear endosperm

Option 2:
Innermost layers of the seed coat

Option 3:
Degenerated nucellus

Option 4:
Immature embryo

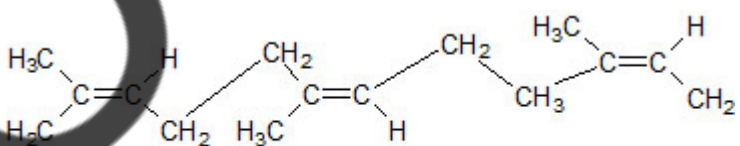
Correct Answer:
Free nuclear endosperm

Solution:
As we learnt in

Edible parts of fruits -

In mango elible part is mesocarp. In ber both epicarp and mesocarp are edible and endocarp is drupe. In almond & walnut, the elible part is seed. In coconut epicarp is hard & fluir, mesocarp is flick & consists fibers. The endocarp is hard & seed is protected in it. The sweet water of coconut is liquid & solid endosperm.

-



(Natural rubber)

Natural rubber is Grinar polymer of isoprene at is called cis-1, 4-polyisoprene.

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Q. 12 The cutting of DNA at specific locations became possible with the discovery of:

Option 1:
Probes

Option 2:
Selectable markers

Option 3:
Ligases

Option 4:
Restriction enzymes

Correct Answer:
Restriction enzymes

Solution:
Restriction enzymes are DNA cutting enzymes found in bacteria. A restriction enzyme recognizes and cuts DNA only at a particular sequence of nucleotides. For example, the bacterium Haemophilus aegyptius produces an enzyme named Hae III that cuts DNA wherever, it encounters the sequence.

Q. 13 Which of the following structures is **not** found in a prokaryotic cell ?

Option 1:
Ribosome

Option 2:
Mesosome

Option 3:
Plasma membrane

Option 4:
Nuclear envelope

Correct Answer:
Nuclear envelope

Solution:
The nuclear membrane is absent in a prokaryotic cell. The nuclear membrane is a double membrane layer which surrounds the nucleus. It is made up of lipids, that encases the genetic material. Hence, the correct answer is 'Nuclear membrane'.

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Q. 14 Arrange the following events of meiosis in correct sequence:

a. Crossing over
b. Synapsis
c. Terminalisation of chiasmata
d. Disappearance of nucleolus

Option 1:
(b),(a),(c),(d)

Option 2:
(a),(b),(c).,(d)

Option 3:
(b),(c),(d),(a)

Option 4:
(b),(a),(d),(c)

Correct Answer:
(b),(a),(c),(d)

Solution:
Correct sequence → (b) Synapsis (a) crossing over (c)Terminalisation of chiasmata (d) Disappearance of nucleolus.
∴b,a,c,d
Synapsis happens in the second stage nucleolus of meiosis I.
crossing happens in the third stage pachytene of meiosis I.
Terminalisation of chiasmata makes the beginning of the final stage of meiosis I. Disappearance of nucleolus happens by the end of final stage Diakinesis of meiosis I.

Q. 15 A column of water within xylem vessels of tall trees does **not** break under its weight because of:

Option 1:
Tensile strength of water

Option 2:
Lignification of xylem vessels

Option 3:
Positive root pressure

Option 4:
Dissolved sugars in water

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Correct Answer:
Tensile strength of water

Solution:
As we learnt in

Cohesion tension or transpirational pull theory -

There is a continuous column of water from root through the stem and into leaves. Water molecules are attached with each other by cohesion force, and by walls of tracheary by adhesion force. Tensile strength of water is very high which keeps the cohesive forces in function. Hence, the correct answer is option a.

Q. 16 The imperfect fungi which are decomposers of litter and help in mineral cycling belong to :

Option 1:
Basidiomycetes

Option 2:
Phycomycetes

Option 3:
Ascomycetes

Option 4:
Deuteromycetes

Correct Answer:
Deuteromycetes

Solution:
As we learnt in

Deuteromycetes / Fungi imperfecti -

1. Mycelium is seperate and branched.
2. Sexual reproduction absent
3. Fruiting body absent.
4. Asexual reproduction by conidia.

- wherein

Eg. Alternaria, Trichoderma.

Q. 17 The structures that help some bacteria to attach to rocks and/or host tissues are :

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Option 1:
Fimbriae

Option 2:
Mesosomes

Option 3:
Holdfast

Option 4:
Rhizoids

Correct Answer:
Fimbriae

Solution:
The features that assist some bacteria to cling to rocks and /or host tissues are fimbriae. Fimbriae are long filament polymeric structural proteins found on the surface of bacterial cells. They make it feasible for the bacteria to bind to certain receptor structures and colonize particular surfaces.

Q. 18 The DNA molecule to which the gene of interest is integrated for cloning is called :

Option 1:
Vector

Option 2:
Template

Option 3:
Carrier

Option 4:
Transformer

Correct Answer:
Vector

Solution:
The DNA molecule to which the gene of interest is integrated for cloning is called vector. It is a DNA molecule used as a vehicle to artifically carry foreign genetic material into another cell, A vector containing foreign DNA is termed as recombinant DNA.

Q. 19 Pick up the **wrong** statement:

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Option 1:
Protista have photosynthetic and heterotrophic modes of nutrition

Option 2:
Some fungi are edible

Option 3:
Nuclear membrane is present in Monera

Option 4:
Cell wall is absent in Animalia

Correct Answer:
Nuclear membrane is present in Monera

Solution:
As we learnt in

Characteristics of kingdom Monera – -

- Prokaryotic organism without nuclear membrane, nucleolus chromatin
- Naked DNA and RNA
- Membrane bound organells absent
- Ribosome : 70 S type
- Reproduction by asexual method
- Cell division is amitotic.

-

Q. 20 Metagenesis refers to :

Option 1:
Alternation of generation between asexual and sexual phases of an organism

Option 2:
Occurrcnce of a drastic change in form during post-embryonic development

Option 3:
Presence of a segmented body and parthenogenetic mode of reproduction

Option 4:
Presence of different morphic forms

Correct Answer:
Alternation of generation between asexual and sexual phases of an organism

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Solution:

Metagenesis in Cnidaria -

Alternation of generation or metagenesis is exhibited.

- wherein

$$Polyp \overset{asexual\,reproduction}{\rightarrow} medusa \overset{sexual\,reproduction}{\rightarrow} polyp$$

In coelenterates, metagenesis is alternation of generation between polyp and medusa. Polyp reproduces asexually by budding to form medusa and medusa reproduces sexually to form polyp.

Q. 21 Which of the following events is not associated with ovulation in human female?

Option 1:
Full development of Graafian follicle

Option 2:
Release of secondary oocyte

Option 3:
LH surge

Option 4:
Decrease in estradiol

Correct Answer:
Decrease in estradiol

Solution:
As we discussed

Ovulatory Phase -

The secretion of gonadotropins (LH and FSH) increases gradually during the follicular phase, and stimulates follicular development as well as secretion of estrogens by the growing follicles. Both LH and FSH attain a peak level in the middle of cycle (about 14th day).

- wherein

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Rapid secretion of LH leading to its maximum level during the mid-cycle called LH surge induces rupture of Graafian follicle and thereby the release of
ovum (ovulation).

During ovaluation graafian follicles get ruptured after full devlopment . Estradoil levels increase.

Q. 22 Which of the following joints would allow no movement ?

Option 1:
Cartilaginous joint

Option 2:
Synovial joint

Option 3:
Ball and Socket joint

Option 4:
Fibrous joint

Correct Answer:
Fibrous joint

Solution:
Due to their immobility, fibrous joints are sometimes known as fixed or immovable joints. Since there is no joint cavity or space between the bones in these joints, they cannot move. Fibrous connective tissue holds them together. Sutures, syndesmoses, and gomphoses are the three different varieties of fibrous joints. Only the skull contains sutures. The ankle's tibia and fibula joint is an example of a syndesmosis. Gomphoses form between sockets and their teeth.

The right response is thus "Fibrous joint."

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Q. 23 Match the following list of microbes and their importance:

(a)	***Sacharomyces***	(i)	Production of immunosuppressive agents
(b)	***Monascus purpureus***	(ii)	Ripening of Swiss cheese
(c)	***Trichoderma polysporum***	(iii)	Commercial production of ethanol
(d)	***Propionibacterium sharmanii***	(iv)	Production of blood-cholesterol lowering agents

(a) (b) (c) (d)

Option 1:
(iv) (iii) (ii) (i)

Option 2:
(iv) (ii) (i) (iii)

Option 3:
(iii) (i) (iv) (ii)

Option 4:
(iii) (iv) (i) (ii)

Correct Answer:
(iii) (iv) (i) (ii)

Solution:

As we have learnt in Chemicals, Enzymes and other Bioactive Molecules -

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Microbes in household products -Roquefort cheese -

The 'Roquefort cheese' are ripened by growing a specific fungi on them, which gives them a particular flavour.

Microbes used in Fermented Beverages -

Microbes especially yeasts have been used for the production of beverages like wine, beer, whisky, brandy or rum.

- wherein

For this purpose the yeast *Saccharomyces cerevisiae*, is used for fermenting malted cereals and fruit juices, to produce ethanol.

cyclosporin A, -

A bioactive molecule, produced by the fungus *Trichoderma polysporum*

- wherein

is used as an immunosuppressive agent in organ-transplant patients

Hence, the correct option is (d).

Q. 24 The UN conference of Parties on climate change in the year 2012 was held at:

Option 1:
Doha

Option 2:
Lima

Option 3:
Warsaw

Option 4:
Durban

Correct Answer:
Doha

Solution:
The UN conference of parties on climate change in the year 2012 was held at Doha.

Q. 25 If you suspect major deficiency of antibodies in a person, to which of the following would you look for confirmatory evidence ?

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Option 1:
Serum albumins

Option 2:
Haemocytes

Option 3:
Serum globulins

Option 4:
Fibrinogin in plasma

Correct Answer:
Serum globulins

Solution:
As we learnt in

Antibodies -

The B-lymphocytes produce an army of proteins in response to pathogens into our blood to fight with them.

- wherein

These proteins are called antibodies.

Globulin is one of the protein found in serum and it includes proteins, enzymes, complement and immunoglobins (antibody).That's why, if major deficiency of antibodies is suspected in a person, the globulins in serum is tested as the confirmatory evidence.

Q. 26 Chromatophores take part in :

Option 1:
Growth

Option 2:
Movement

Option 3:
Respiration

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Option 4:
Photosynthesis

Correct Answer:
Photosynthesis

Solution:
- **Chromatophores are involved in photosynthesis.**
- Members of phototrophic bacteria contain chromatophores.
- They participate in photosynthesis and contain carotenoids and bacteriochlorophyll pigments.
- The chromatophore membranes are made entirely of the light-harvesting proteins in purple bacteria like *Rhodospirillum rubrum.*
- Green sulphur bacteria, on the other hand, are arranged in chlorosomes, which are specialized antenna complexes.

Hence, the correct answer is **option 4.**

Q. 27 Acid rain is caused by increase in the atmospheric concentration of:

Option 1:
SO_3 and CO

Option 2:
CO_2 and CO

Option 3:
CO_3 and dust

Option 4:
SO_2 and NO_2

Correct Answer:
SO_2 and NO_2

Solution:
Acid rains are caused due to increase in the concentration of SO_2 and NO_2. SO_2 form sulphuric acid while NO_2 form nitric acid.

Q. 28 During ecological succession :

Option 1:
the establishment of a new biotic community is very fast in its primary phase.

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Option 2:
the numbers and tvpes of animals remain constant.

Option 3:
the changes lead to a community that is in near equilibrium with the environment and is called pioneer community

Option 4:
the gradual and predictable change in species composition occurs in a given area

Correct Answer:
the gradual and predictable change in species composition occurs in a given area

Solution:

As discussed in Ecological Succession -

The gradual and fairly predictable change in the species composition of a given area is called ecological succession.

- wherein

During succession some species colonise an area and their populations become more numerous, whereas populations of other species decline and even disappear.

Hence, the correct option is (d).

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Q. 29 The oxygen evolved during photosynthesis comes from water molecules. Which one of the following pairs of elements is involved in this reaction ?

Option 1:
Manganese and Potassium

Option 2:
Magnesium and Molybdenum

Option 3:
Magnesium and Chlorine

Option 4:
Manganese and Chlorine

Correct Answer:
Manganese and Chlorine

Solution:
As we learnt in

Photolysis of water -

Occurs in the lumen of thylakoid. Requires Mn^{+}, Cl^{-}, Ca^{2+} and oxygen evolving complex

- wherein

$$2H_2O \rightarrow 4H^{+} + 4e^{-} + O_2$$

e^{-} move towards P_{680} of PSII and H^{+} accumulate in lumen and finally reduced to NADP

Photolysis of water requires Mn^{2+}, Cl^{-} and Ca^{2+}

Q. 30 Which of the following pairs is **not** correctly matched ?

	Mode of reproduction	**Example**

Option 1:
Rhizome Banana

Option 2:
Binary fission *Sargassum*

Option 3:
Conidia *Penicillium*

Option 4:
Offset Water hyacinth

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Correct Answer:
Binary fission *Sargassum*

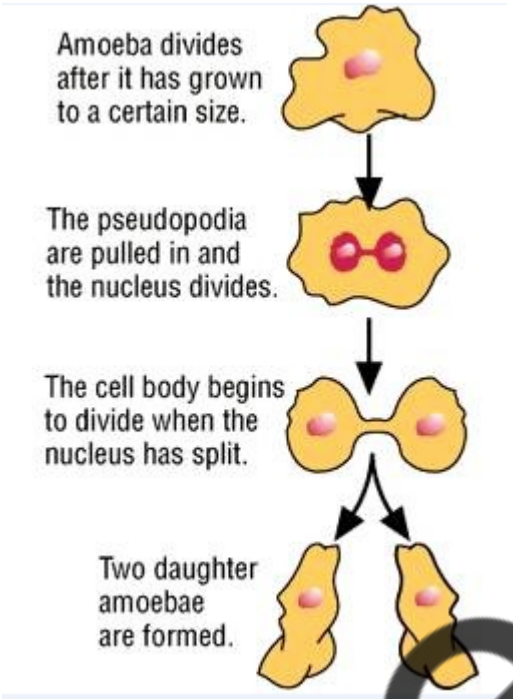
Solution:
As we discussed in

Binary fission -

Many single-celled organisms reproduce by binary fission, where a cell divides into two halves and each rapidly grows into an adult.

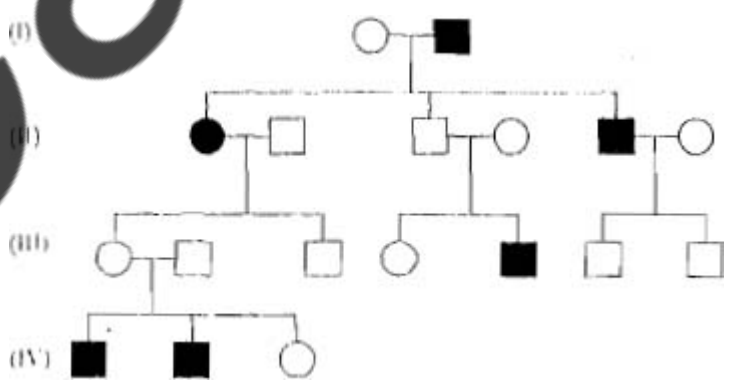
- wherein

e.g.,Amoeba, Paramecium.



Binary fission generally occurs in single celled organisms. Sargassum is a diploid sporophyte in which vegetative reproduction is through fragmentation.

Q. 31 In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.



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Option 1:
X-linked recessive

Option 2:
Autosomal recessive

Option 3:
X-linked dominant

Option 4:
Autosomal dominant

Correct Answer:
Autosomal recessive

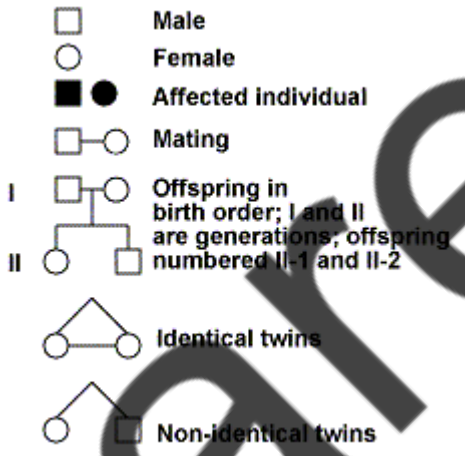
Solution:
As we learnt in

Pedigree analysis in human genetics -

In human genetics, the pedigree study provides a strong tool, which is utilised to trace the inheritance of a specific trait, abnormality or disease.

- wherein

Some of the important symbols used in the pedigree analysis are shown.



The given pedigree shows the autosomal recessive disorder. In this disorder, the individual inherit two mutated genes, one from each parent. This disorder is usually passed on by two carriers. Health is rarely affected, but individual have one mutated gene(recessive gene) and one normal gene(dominant gene) for the condition. The carriers have a 25% chance of having an unaffected child with normal genes, 50% chance of having an unaffected child who also is a carrier and a 25% chance of having an affected chid with recessive genes.

Q. 32 Which one of the following animals has two separate circulatory pathways ?

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Option 1:
Lizard

Option 2:
Whale

Option 3:
Shark

Option 4:

Frog

Correct Answer:
Whale

Solution:
As we learnt in

Heart in mammalia -

4-chambered, two auricles and two ventricles.

-

Whale is a mammal which has 4 chambered heart, so has complete separation of oxygenated and deoxygenated blood. Whales have double circulatory pathways: Systemic and pulmonary circulation

---

Q. 33 Flowers are unisexual in .

Option 1:
Cucumber

Option 2:
China rose

Option 3:
Onion

Option 4:
Pea

Correct Answer:
Cucumber

Solution:
As discussed in

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Unisexual flower -

Only one either androecium (staminate) or gynoecium (Pistillate) present in one flower

-

Unisexual flowers are those in which only one sex organ is present. E.g. cucumber.

Q. 34 Which one of the following fruits is parthenocarpic?

Option 1:
Apple

Option 2:
Jackfruit

Option 3:
Banana

Option 4:
Brinjal

Correct Answer:
Banana

Solution:
As we learnt in

Fruits -

Fruit is matured or ripned ovary developed after fertilization. Afetr fertilization ovary forms fruits & ovules form the seeds.

- wherein

Pomology is the study of fruits.

Some fruit do not contain seeds called pardhenocarpic or seedless fruits ex. banana, grapes, pineapple (fruits without fertilization)

Banana is parthenocarpic or seedless fruit.

Q. 35 A pleiotropic gene :

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Option 1:
is a gene evolved during Pliocene.

Option 2:
controls a trait only in combination with another gene

Option 3:
controls multiple traits in an individual.

Option 4:
is expressed only in primitive plants.

Correct Answer:
controls multiple traits in an individual.

Solution:
A pleiotropic gene is one that regulates several phenotypic features in a person. A pleiotropic gene influences metabolic pathways, which result in a variety of phenotypes. Pleiotropism is demonstrated by phenylketonuria. A mutation in the gene that encodes for phenylalanine hydroxylase causes this condition. Phenylketonuria is distinguished by phenotypic characteristics such as skin coloring, hair loss, and mental impairment.

Q. 36 Which of the following is **not** a function of the skeletal system ?

Option 1:
Storage of minerals

Option 2:
Production of body heat

Option 3:
Locomotion

Option 4:
Production of erythrocytes

Correct Answer:
Production of body heat

Solution:
As we learnt in

SKELETAL SYSTEM -

Skeletal system consists of a framework of bones and a few cartilages.

- wherein

This system has a significant role in movement shown by the body.

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Production of Heat is not by skeleton system. It provides support for movement, store minerals $(C_a{}^{2+})$ and production of blood cells(Bonemarrow).

Q. 37 A jawless fish, which lays eggs in fresh water and whose ammocoetes larvae after metamorphosis return to the ocean is :

Option 1:
Myxine

Option 2:
Neomyxine

Option 3:
Petromyzon

Option 4:
Eptatretus

Correct Answer:
Petromyzon

Solution:
As we learnt in

Class Cyclostomata -

All are ectoparasites on some fishes. Mouth is circular and jawless. Head and brain poorly developed.

- wherein

Eg. Petromyzon, Myxine.

Petromyzon(the lamprey) belongs to the section Agnatha of the sub-phylum-Vertebrata. They have long, greenish brown, cylindrical body with smooth scaleless, slimy skin, jawless mouth, etc. They lay eggs in freshwater but their ammocoete larvae(lower) after metamorphosis return to the ocean.

Q. 38 Filiform apparatus is characteristic feature of:

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Option 1:
Nucellar embryo

Option 2:
Aleurone cell

Option 3:
Synergids

Option 4:
Generative cell

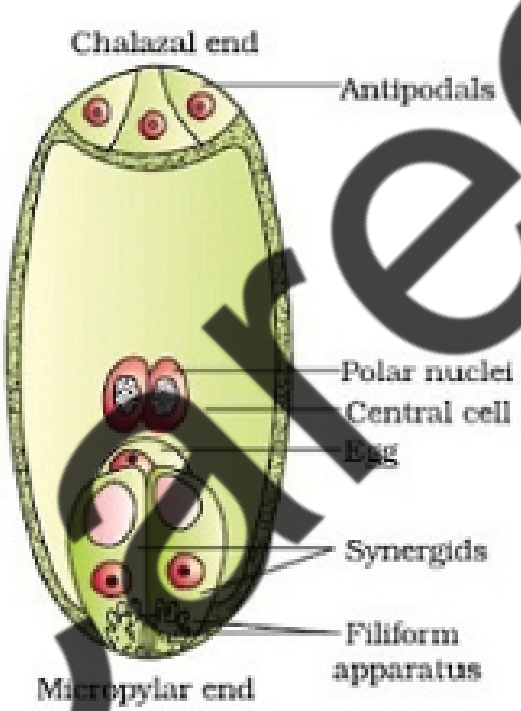
Correct Answer:
Synergids

Solution:
As we learnt in

Structure of embryo sac -

The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. A typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled

- wherein



Filiform apparatus is cellular thickening at the micropylar tip of synergids.

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Q. 39 Read the different components from (a) to (d) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem :

a. Secondary cortex
b. Wood
c. Secondary phloem
d. Phellem

The correct order is :

Option 1:
(a),(b),(d),(c)

Option 2:
(d),(a),(c),(b)

Option 3:
(d),(c),(a),(b)

Option 4:
(c),(d),(b),(a)

Correct Answer:
(d),(a),(c),(b)

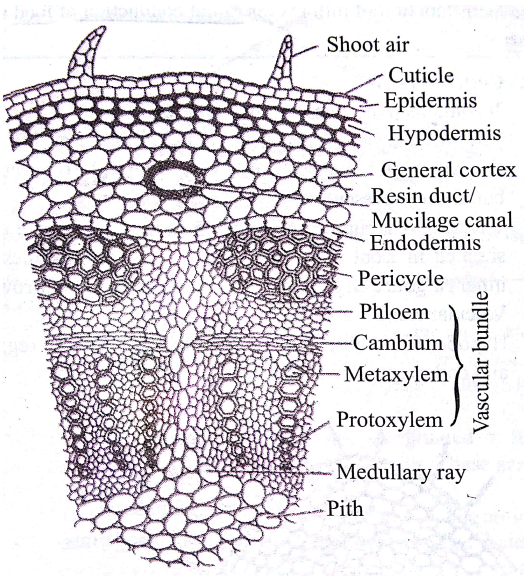
Solution:
As we learnt in

Anatomy of Dicot Stems -

- In contains outer single layer called epidermis, convered by cuticle, bears trichomes (multicellular hairs). Epidermis lacks chloroplast.
- Cortex is divided into hypodermis, general cortex & endodermis.
- Pericycle is situated in between endodermis & vascular bundles.
- Vascular bundles (wedge shaped) are arranged in a ring. It is conjoint, collateral and open. Eustele is present.
- Pith is made up of paranchyma cells. Function of pith is storage of water and food.
- Pith ray or medullary ray is present built in pith which radially conduct food and water.

- wherein

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Annual rings (wood) is present between medullary rays and pith.

Q. 40 Which one of the following hormones is **not** involved in sugar metabolism ?

Option 1:
Aldosterone

Option 2:
Insulin

Option 3:
Glucagon

Option 4:
Cortisone

Correct Answer:
Aldosterone

Solution:
As we learnt in

Cortisol -

In our body, cortisol is the main glucocorticoid.

- wherein

Corticoids, which regulate the balance of water and electrolytes in our body are called mineralocorticoids. Aldosterone is the main mineralocorticoid in our body.

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Aldosterone -

Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na+ and water and excretion of K+ and phosphate ions.

- wherein

Thus, aldosterone helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure. Small amounts of androgenic steroids are also secreted by the adrenal cortex which play a role in the growth of axial hair, pubic hair and facial hair during puberty

Aldosterone regulates balance of water electrolyte in body.

Q. 41 Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of:

Option 1:
Vitamin C

Option 2:
Omega 3

Option 3:
Vitamin A

Option 4:
Vitamin B

Correct Answer:
Vitamin A

Solution:
As we learnt in

Methods of breeding for disease resistance crops -

Genetic Engineering

- wherein

Genes expected to confer disease resistance are isolated, cloned and transferred into the crop in question. In case of viral pathogens, several transgenes have been evaluated, viz, virus coat protein gene, DNA copy of viral satellite RNA, defective viral genome, antisense constructs of critical viral genes, and ribozymes. Viral coat protein gene approach seems to be the most successful. A virus transgenic variety of squash is in commercial cultivation in U.S.A

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Golden rice is a variety of rice produced through genetic engineering to biosynthesize beta-carotene, a precursor of vitamin A, in the edible parts of rice. It is intended to produce a fortified food to be grown and consumed in areas with a shortage of dietary vitamin A.

Correct option is 3.

Q. 42 Outbreeding is an important strategy of animal husbandry because it:

Option 1:
is useful in producing purelines of animals.

Option 2:
is useful in overcoming inbreeding depression.

Option 3:
exposes harmful recessive genes that are eliminated by selection.

Option 4:
helps in accumulation of superior genes.

Correct Answer:
is useful in overcoming inbreeding depression.

Solution:

As we learnt in

Out-crossing -

This is the practice of mating of animals within the same breed, but having no common ancestors on either side of their pedigree up to 4-6 generations.

- wherein

The offspring of such a mating is known as an out-cross. It is the best breeding method for animals that are below average in productivity in milk production, growth rate in beef cattle, etc. A single outcross often helps to overcome inbreeding depression.

It is the best breeding method for animals that are below average in productivity in milk production, growth rate in beef cattle, etc. A single outcross often helps to overcome inbreeding depression.

Q. 43 A gene showing codominance has :

Option 1:
alleles tightly linked on the same chromosome

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Option 2:
alleles that are recessive to each other

Option 3:
both alleles independently expressed in the heterozygote

Option 4:
one allele dominant on the other

Correct Answer:
both alleles independently expressed in the heterozygote

Solution:
As we learnt in

Co-dominance explained in red-blood cells -

Red-blood cells are controlled by gene (I). RBC's have sugar attached to surface which is controlled by the gene (I). The gene (I) has three alleles : I^A , I^B and i

- wherein

The allele of gene (I), I^A and I^B process sugar of different forms while i does not produce any.

A gene shows codominance when both alleles in heterozygous condition, express their traits independently instead of showing dominant-recessive relationship and such alleles are called codominant alleles.

Q. 44 Which one of the following hormones though synthesised elsewhere, is stored and released by the master gland?

Option 1:
Luteinizing hormone

Option 2:
Prolactin

Option 3:
Melanocyte stimulating hormone

Option 4:
Antidiuretic hormone

Correct Answer:
Antidiuretic hormone

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Solution:
As we learnt in

Vasopressin -

Vasopressin acts mainly at the kidney and stimulates resorption of water and electrolytes by the distal tubules and thereby reduces loss of water through urine (diuresis)

- wherein

Hence, it is also called as anti-diuretic hormone (ADH).

Osmoreceptors -

Osmoreceptors in the body are activated by changes in blood volume, body fluid volume and ionic concentration.

- wherein

An excessive loss of fluid from the body can activate these receptors which stimulate the hypothalamus to release antidiuretic hormone (ADH) or vasopressin from the neurohypophysis.

Q. 45 Increase in concentration of the toxicant at successive trophic levels is known as :

Option 1:
Biodeterioration

Option 2:
Biotransformation

Option 3:
Biogeochemical cycling

Option 4:
Biomagnification

Correct Answer:
Biomagnification

Solution:

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As discussed in Water Pollution & Its Causes : Industrial Effluents & Biomagnification -

Biomagnification -

Biomagnification refers to increase in concentration of the toxicant at successive trophic levels. This happens because a toxic substance accumulated by an organism cannot be metabolised or excreted, and is thus passed on to the next higher trophic level.

- wherein

This phenomenon is wellknown for mercury and DDT.

Hence, the correct option is (d).

Q. 46 Industrial melanism is an example of:

Option 1:
Natural selection

Option 2:
Mutation

Option 3:
Neo Lamarckism

Option 4:
Neo Darwinism

Correct Answer:
Natural selection

Solution:
As we learnt in

Industrial Melanism -

It is the appearance of dark melanic forms of some organism like moths in the industrial region where soot emission from burning of coal has been heavy so as to color the background.

- wherein

Example of the moth that survived natural selection as it was black and escaped unnoticed by the birds.

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Q. 47 The primary dentition in human differs from permanent dentition in **not** having one of the following type of teeth :

Option 1:
Bicuspids

Option 2:
Molars

Option 3:
Incisors

Option 4:
Cuspids

Correct Answer:
Bicuspids

Solution:
As we learnt in

Decidous Teeth -

In humans, first set of teeth are called the decodous teeth, which are 20 in number

- wherein

The dental formula is 2102/2102.

This means 2 incisors, 1 canine, 0 premolars, 2 molars

Premolars are absent 2102/2102

Premolars are known as Bicuspids and Canines are Cuspids

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Q. 48 The wheat grain has an embryo with one large, shield-shaped cotyledon known as :

Option 1:
Coleorrhiza

Option 2:
Scutellum

Option 3:
Coleoptile

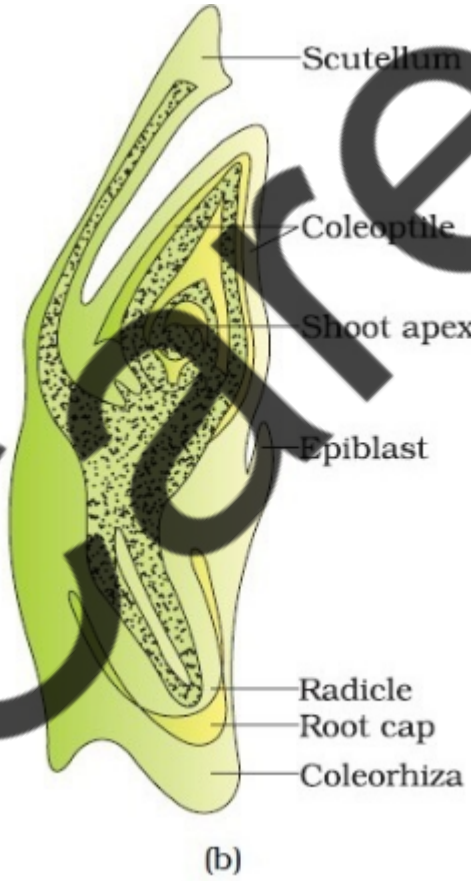
Option 4:
Epiblast

Correct Answer:
Scutellum

Solution:
As we learnt in

Monocotyledons embryo -

Embryos of monocotyledons possess only one cotyledon. In the grass family the cotyledon is called scutellum that is situated towards one side (lateral) of the embryonal axis. At its lower end, the embryonal axis has the radical and root cap enclosed in an undifferentiated sheath called coleorrhiza.



(b)

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In monocots grass family cotyledon is called scutellum. wheat grain is monocot belonging to grass family. Hence, the correct answer is option b.

Q. 49 The body cells in cockroach discharge their nitrogenous waste in the haemolymph mainly in the form of :

Option 1:
Potassium urate

Option 2:
Urea

Option 3:
Calcium carbonate

Option 4:
Ammonia

Correct Answer:
Potassium urate

Solution:
As we learnt in

Composition of hemocoel -

Fat boides are present in hemocoel which are trophocytes, mycetocytes, oenocyte and Urate cells.

- wherein

Fat bodies of cockroach is analogous to liver of vertebrate. The body cell discharges their N_2 waste in the hemolymph in the form of potassium urate

Q. 50 Which of the following biomolecules does have a phosphodiester bond ?

Option 1:
Monosaccharides in a polysaccharide

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Option 2:
Amino acids in a polypeptide

Option 3:
Nucleic acids in a nucleotide

Option 4:
Fatty acids in a diglyceride

Correct Answer:
Nucleic acids in a nucleotide

Solution:

As we learnt in

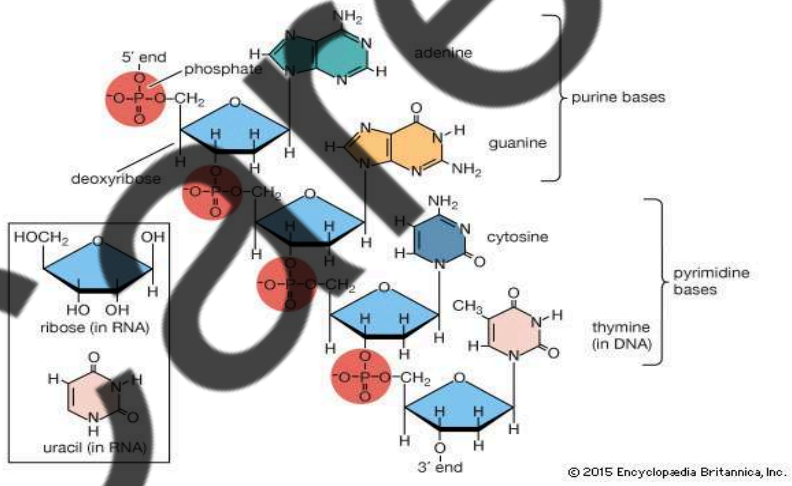
Phosphodiester Bond - The bond between a phosphate and hydroxyl group of sugar form an ester linkage, it is called phosphodiester bond.

-

Nucleic Acids -

Nucleic Acids like DNA, RNA contains only Nucleotides and function as genetic material.

- wherein



Nucleic acid are polymer of nucleotides joined by phosphodiester bonds.

Q. 51 The term "linkage" was coined by :

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Option 1:
T. Boveri

Option 2:
G. Mendel

Option 3:
W. Sutton

Option 4:
T.H. Morgan

Correct Answer:
T.H. Morgan

Solution:
As we learnt in

Linkage -

Morgan experimented and found that when the two genes in a hybrid cross situated on same chromosone, the proportion of parental gene combination were much higher than the non-parental type.

- wherein

Morgan attributes this to the physical association or linkage of the two genes and called it linkage.

Q. 52 Which one is a **wrong** statement ?

Option 1:
Mucor has biflagellate zoospores

Option 2:
Haploid endosperm is typical feature of gymnosperms

Option 3:
Brown algae have chlorophyll a and c, and fucoxanthin

Option 4:
Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms

Correct Answer:
Mucor has biflagellate zoospores

Solution:

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The wrong statement is:

Option 1) Mucor has biflagellate zoospores.

This statement is incorrect. Mucor is a type of fungi that reproduces asexually through the production of asexual spores called sporangiospores. These spores are non-motile and are not flagellated.

Q. 53 Ectopic pregnancies are referred to as:

Option 1:
Implantation of embryo at site other than uterus.

Option 2:
Implantation of defective embryo in the uterus

Option 3:
Pregnancies terminated due to hormonal imbalance.

Option 4:
Pregnancies with genetic abnormality.

Correct Answer:
Implantation of embryo at site other than uterus.

Solution:
Ectopic pregnancy refers to the situation in which the fertilised egg implants itself outside of the uterus. The fertilised egg passes through the fallopian tube and adheres to the uterine lining normally. Ectopic pregnancy typically happens when the egg is placed in one of the fallopian tubes. So, it is also known as a tubal pregnancy. The fallopian tube is unable to support the expanding uterus, which could cause serious issues with foetal development. Therefore, "Implantation of embryo at site other than uterus" is the right response.

Q. 54 Most animals that live in deep oceanic waters are :

Option 1:
secondary consumers

Option 2:
tertiary consumers

Option 3:
detritivores

Option 4:
primary consumers

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Correct Answer:
detritivores

Solution:
As we learnt in

Fragmentation -

Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation.

-

Detritivores are found in the bottom of sea, oceans or lakes. They are called benthos (bottom dewellers).

Q. 55 Which of the following diseases is caused by a protozoan ?

Option 1:
Influenza

Option 2:
Babesiosis

Option 3:
Blastomycosis

Option 4:
Syphilis

Correct Answer:
Babesiosis

Solution:
Babesiosis is a malaria-like parasitic disease caused by infection with Babesia bigemina, a genus of protozoa piroplasms.

Q. 56 In which of the following interactions both partners are adversely affected ?

Option 1:
Predation

Option 2:
Parasitism

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Option 3:
Mutualism

Option 4:
Competition

Correct Answer:
Competition

Solution:

Competition -

- Competition can be best defined as a process in which the fitness of one species (measured in terms of its 'r' the intrinsic rate of increase) is significantly lower in the presence of another species.
- It is a type of interaction, where both the species suffer.

-

Q. 57 Identify the **correct** order of organisation of genetic material from largest to smallest:

Option 1:
Genome, chromosome, nucleotide, gene

Option 2:
Genome, chromosome, gene, nucleotide

Option 3:
Chromosome, genome, nucleotide, gene

Option 4:
Chromosome, gene, genome, nucleotide

Correct Answer:
Genome, chromosome, gene, nucleotide

Solution:
As we learnt in

Chromosomes -

Structure in nucleus that appeared to double and divide just before each cell division.

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- wherein

Discovered by de Vries, correns and Von Tschermark

Feature of Human genome -

Repetitive sequence are stretches of DNA sequence that are repeated many times, sometimes hundred to thousand times. They are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.

-

Structure of polynucleotide chain -

A nucleotide has 3 components:

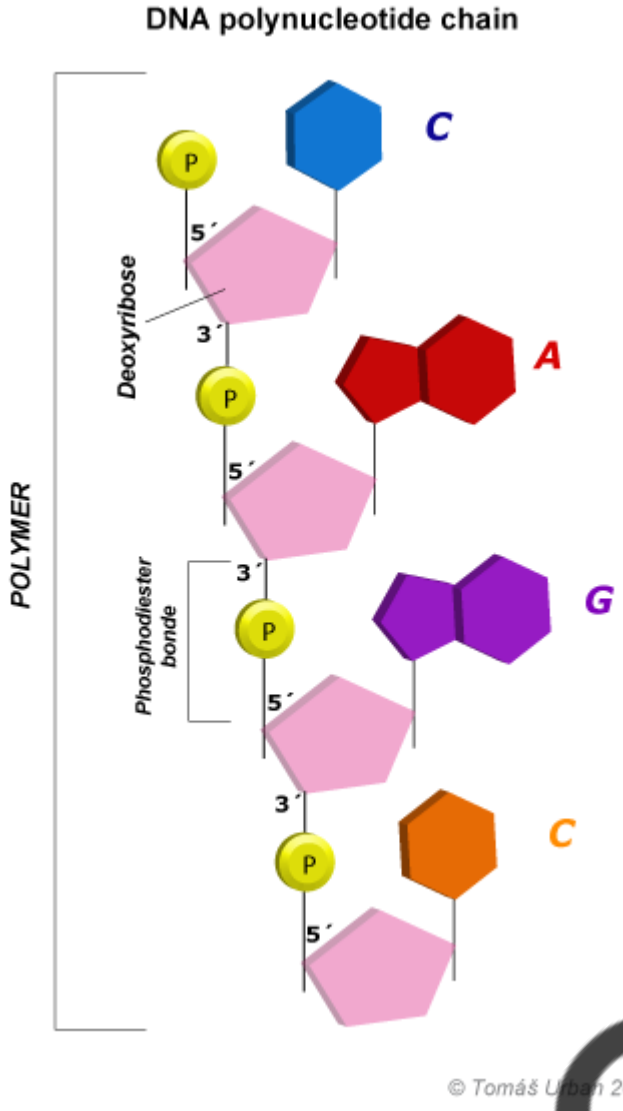
1. A nitrogenous base

2. A pentose sugar

3. And a phosphate group

- wherein

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Q. 58 A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind ?

Option 1:
1

Option 2:
Nil

Option 3:
0.25

Option 4:
0.5

Correct Answer:
0.25

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Solution:
As we learnt in

Explanation of law of independent assortment -

Gametes have gene R and r, and they also have alleles Y and y, the segregation of 50% of R and 50% of r is independent from segregation of 50% Y and 50% of y

- wherein

Thus, there are 4 genotypes of gametes, RY, Ry , rY and ry each with frequency 25% of the total gametes produced.

When a colour blind (XcY) marries a woman with normal sight (XX) who has no family history of colour blindness, all of their sons will be normal pure and all of their daughters will be carriers.

So in the next generation, the children of all of their son will be normal in all conditions(except the case in which the wife involved is not carrier neither colour blind). For carrier daughters:

(i) If they marry a normal man, 50% of their grandsons will become colour blind

(ii) If carrier daughter marries a colour blind man 50% of their grandson will be colour blind along with 50% of the granddaughter while rest 50% of the granddaughters will be carriers.

So in both the above cases, the result shows 50% of grandsons will be colour blind which in terms of overall progress (son + daughter) comes as 25% thus confirming the probability as 0.25.

Correct option is 3.

Q. 59 In photosynthesis, the light-independent reactions take place at :

Option 1:
Photosystem I

Option 2:
Photosystem II

Option 3:
Stromal matrix

Option 4:
Thylakoid lumen

Correct Answer:
Stromal matrix

Solution:

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Dark reaction of photosynthesis occurs in the stroma. Hence, the correct answer is option c.

Q. 60 In which of the following both pairs have **correct** combination ?

Option 1:

Gaseous nutrient cycle	Carbon and sulphur
Sedimentary nutrient cycle	Nitrogen and Phosphorus

Option 2:

Gaseous nutrient cycle	Nitrogen and sulphur
Sedimentary nutrient cycle	Carbon and Phosphorus

Option 3:

Gaseous nutrient cycle	Sulphur and Phosphorus
Sedimentary nutrient cycle	Carbon and Nitrogen

Option 4:

Gaseous nutrient cycle	Carbon and Nitrogen
Sedimentary nutrient cycle	Sulphur and Phosphorus

Correct Answer:

Gaseous nutrient cycle	Carbon and Nitrogen
Sedimentary nutrient cycle	Sulphur and Phosphorus

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Solution:

- The materials that circulate between the biotic and abiotic components of the biosphere are gases or vapours in gaseous **nutrient cycles,** and the reservoir pool is the atmosphere or hydrosphere, such as water, carbon, hydrogen, oxygen, or nitrogen.
- Sulphur is subject to both gaseous and sedimentary nutrient cycles.

There are two types of nutrient cycling:

(i) Gaseous cycles, such as the C and N cycles,

(ii) Sedimentary cycles, such as the P and S cycles,

- Sedimentary cycles involve the cyclic exchange of minerals from the lithosphere or crust of the earth with the biotic components of the ecosystem.
- **Sulphur and phosphorus cycles** are two examples of cycles that fall under the **sedimentary category.**
- Minerals move back and forth between the atmosphere and the biotic world in gaseous cycles.
- **Carbon and nitrogen** cycles are frequent examples of **gaseous cycles.**
- In an ecosystem, a variety of nutrient cycles recycle essential nutrients. **Biogeochemical cycles** are another name for the cycles. The carbon, oxygen, and nitrogen cycles are examples of gaseous nutrient cycles. In these cycles, the component is delivered in the air.
- Sedimentary nutrient cycles incorporate the sulphur cycle and the phosphorus cycle. The element is released into the sedimentary rocks during these cycles.

Hence, the correct solution is **option 4).**

Q. 61 The introduction of t-DNA into plants involves :

Option 1:
Altering the pH of the soil, then heat-shocking the plants

Option 2:
Exposing the plants to cold for a brief period

Option 3:
Allowing the plant roots to stand in water

Option 4:
Infection of the plant by *Agrobacterium tumefaciens*

Correct Answer:
Infection of the plant by *Agrobacterium tumefaciens*

Solution:
By introducing t-DNA in plants, infection of the plant takes place by Agrobacterium tumefaciens. Other factors such as water, cold, and pH do not affect t-DNA.

Q. 62 The wings of a bird and the wings of an insect are :

Option 1:
analogous structures and represent convergent evolution

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Option 2:
phylogenetic structures and represent divergent evolution

Option 3:
homologous structures and represent convergent evolution

Option 4:
homologous structures and represent divergent evolution

Correct Answer:
analogous structures and represent convergent evolution

Solution:
As we learnt in

Analogous Organs -

The organs which have the same function and superficially alike but are quite different in fundamental structure and embryonic origin.

- wherein

Eg: insect and bird wings, Dragonfly, pterodactyl, Eagle, humorous, bat.

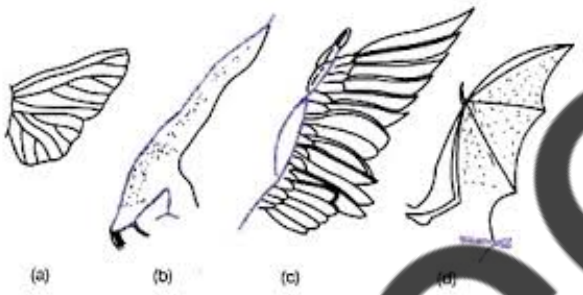


Fig. 7.4 Examples of analogous organs (a) Wing of insect (b) Wing of *Pterodactyl* (c) Wing of bird (d) Wing of bat

The organs which have the same function and superficially alike but are quite different in fundamental structure and embryonic origin.

Q. 63 Root pressure develops due to :

Option 1:
Low osmotic potential in soil

Option 2:
Passive absorption

Option 3:
Increase in transpiration

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Option 4:
Active absorption

Correct Answer:
Active absorption

Solution:
As we learnt in

Root pressure theory -

Due to movement of water from soil into the root hair and from there to xylem cells, a hydrostatic root pressure develops which pushes the water up in the xylem vessels.

- wherein

Given by Priestly (1916), the magnitude of pressure is too small for tall trees, hence present in only herbaceous plants.

Hydrostatic root pressure develop in the plant roots by active absorption of nutrients from soil.

Q. 64 Human urine is usually acidic because :

Option 1:
excreted plasma proteins are acidic.

Option 2:
potassium and sodium exchange generates acidity.

Option 3:
hydrogen ions are actively secreted into the filtrate.

Option 4:
the sodium transporter exchanges one hydrogen ion for each sodium ion, in peritubular capillaries.

Correct Answer:
hydrogen ions are actively secreted into the filtrate.

Solution:
As we learnt in

Characteristics of urine -

An adult human excretes, on an average, 1 to 1.5 litres of urine per day.

- wherein

The urine formed is a light yellow coloured watery fluid which is slightly acidic (pH-6.0) and has a characterestic odour.

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The pH of urine is 6.0 which is acidic due to secretion of H^+ ions

H^+ is proportional to 1/pH (low pH -> High H^+)

Q. 65 A protoplast is a cell:

Option 1:
without nucleus

Option 2:
undergoing division

Option 3:
without cell wall

Option 4:
without plasma membrane

Correct Answer:
without cell wall

Solution:
Protoplasts are cells with the plasma membrane, cytoplasm and nucleus, which have had their cell wall removed by the action of enzymes. Cellulase enzymes digest the cellulose in plant cell walls while pectinase enzymes break down the pectin holding cells together.

Q. 66 The species confined to a particular region and not found elsewhere is termed as :

Option 1:
Alien

Option 2:
Endemic

Option 3:
Rare

Option 4:
Keystone

Correct Answer:
Endemic

Solution:

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As we learnt in

Endemism -

Endemism is the ecological state of a species being unique to a defined geographic location, such as an island, nation, country or other defined zone, or habitat type;

- wherein

organisms that are indigenous to a place are not endemic to it if they are also found elsewhere.

Q. 67 Select the **wrong** statement:

Option 1:
W.M. Stanley showed that viruses could be crystallized

Option 2:
The term *'contagium vivum fluidum'* was coined by M. W. Beijerinek

Option 3:
Mosaic disease in tobacco and AIDS in human being are caused by viruses

Option 4:
The viroids were discovered by D.J. Ivanowski

Correct Answer:
The viroids were discovered by D.J. Ivanowski

Solution:
Viroids have been found and designated by Theodor Otto Diener in the year 1971, a plant pathologist who worked at the Agricultural Research Center in Maryland. The Potato spindle tuber viroid (PsTVd) was the first viroid found, and it causes disease in potatoes and other Solanaceae plants. The virus was discovered by D.J. Ivanowski.

Q. 68 Axile placentation is present in :

Option 1:
Lemon

Option 2:
Pea

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Option 3:
Argemone

Option 4:
Dianthus

Correct Answer:
Lemon

Solution:
In axile placentation, the ovary is sectioned by radial spokes with placentas in separate locules. Ovules are arranged along the central axis of placenta. Number of chambers correspond to the number of carpels. It is found in bi or multicarpellary and multilocular ovaries. Examples include China rose, lemon, cotton and tomato.

Q. 69 A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is:

Option 1:
Gamete intra fallopian transfer

Option 2:
Gamete internal fertilization and transfer

Option 3:
Germ cell internal fallopian transfer

Option 4:
Gamete inseminated fallopian transfer

Correct Answer:
Gamete intra fallopian transfer

Solution:
As we learnt in

GIFT (gamete intra fallopian transfer) -

This method involves transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce one, but can provide suitable environment for fertilisation and further development.

-

GIFT is gamete intra fallopian transfer.

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Q. 70 Destruction of the anterior horn cells of the spinal cord would result in loss of :

Option 1:
voluntary motor impulses

Option 2:
commissural impulses

Option 3:
integrating impulses

Option 4:
sensory impulses

Correct Answer:
voluntary motor impulses

Solution:
Destruction of the anterior horn cells of the spinal cord would result in loss of voluntary motor impulses. It is because the anterior horn cells (grey column), contains motor neurons that affect the axial muscles.

Q. 71 During biological nitrogen fixation, inactivation of nitrogenase by oxygen poisoning is prevented by:

Option 1:
Xanthophyll

Option 2:
Carotene

Option 3:
Cytochrome

Option 4:
Leghaemoglobin

Correct Answer:
Leghaemoglobin

Solution:
Leghemoglobin is an oxygen scavenger and protects nitrogenase from O_2

Q. 72 An association of individuals of different species living in the same habitat and having functional interactions is :

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Option 1:
Biotic community

Option 2:
Ecosystem

Option 3:
Population

Option 4:
Ecological niche

Correct Answer:
Biotic community

Solution:
As we learnt in

primary succession -

Examples of areas where primary succession occurs are newly cooled lava, bare rock, newly created pond or reservoir.

- wherein

The establishment of a new biotic community is generally slow. Before a biotic community of diverse organisms can become established, there must be soil. Depending mostly on the climate, it takes natural processes several hundred to several thousand years to produce fertile soil on bare rock.

An association of individual of different species living in the same habitat and having functional interaction is called biotic community. The biotic community is dominated by one of the character. It determines the nature of community. It can be biotic or abioitic.

Q. 73 Name the pulmonary disease in which alveolar surface area involved in gas exchange is drastically reduced due to damage in the alveolar walls.

Option 1:
Emphysema

Option 2:
Pneumonia

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Option 3:
Asthma

Option 4:
Pleurisy

Correct Answer:
Emphysema

Solution:
Emphysema is a long-term condition in which the breathing surface is significantly reduced due to damage to the walls of the alveoli. Hence, option 1 is correct.

Q. 74 Balbiani rings are sites of :

Option 1:
Nucleotide synthesis

Option 2:
Polysaccharide synthesis

Option 3:
RNA and protein synthesis

Option 4:
Lipid synthesis

Correct Answer:
RNA and protein synthesis

Solution:
Balbiani rings are a large puff of polytene chromosomes present during larval development in Dipterians. A gene or set of genes that are actively synthesising RNA is represented by the puffs. The RNA is transported to the cytoplasm to carry out protein synthesis. Only RNA is synthesised in balbiani rings.Option 3 is correct.

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Q. 75 Match the columns and identify the correct option.

	Column I		Column II
(a)	Thylakoids	(i)	Disc-shaped sacs in Golgi apparatus
(b)	Cristae	(ii)	Condensed structure of DNA
(c)	Cisternae	(iii)	Flat membranous sacs in stroma
(d)	Chromatin	(iv)	Infoldings in mitochondria

(a) (b) (c) (d)

Option 1:
(iii) (iv) (i) (ii)

Option 2:
(iii) (i) (iv) (ii)

Option 3:
(iii) (iv) (ii) (i)

Option 4:
(iv) (iii) (i) (ii)

Correct Answer:
(iii) (iv) (i) (ii)

Solution:
As we learnt in

Structure of Nucleus -

The interphase nucleus (nucleus of a cell when it is not dividing) has highly extended and elaborate nucleoprotein fibres called chromatin, nucleus matrix and one or more spherical bodies called nucleoli. The nuclear matrix or the nucleoplasm contains nucleolus and chromatin.

- wherein

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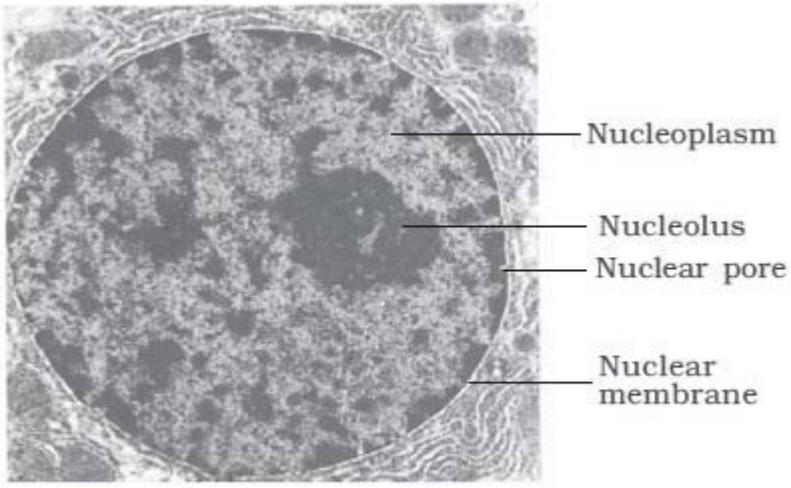


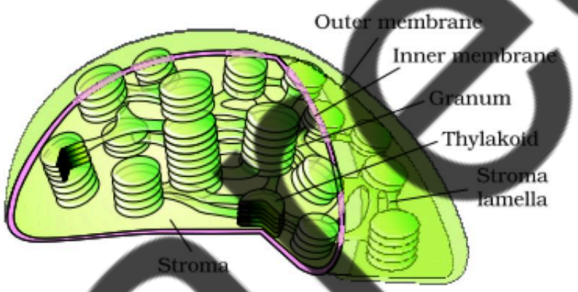
Figure 8.10 Structure of nucleus

And,

Chloroplasts -

Chloroplasts contains chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis.

- wherein



And,

Types of Plastids -

Plastids bear some specific pigments, thus imparting colours to the plants. Base on the type of pigments, plastids can be classified into:

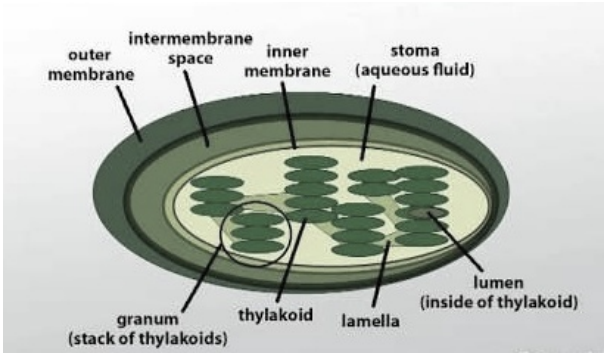
1. Choloroplats

2. Chromoplasts

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3. Leucoplasts

- wherein



And finally,

Golgi Apparatus -

Principally performs the function of packaging materials, to be delivered either to intracellular targets or secreted outside of the cell.

- wherein

It is the important site for the formation of glycoproteins and glycolipids.

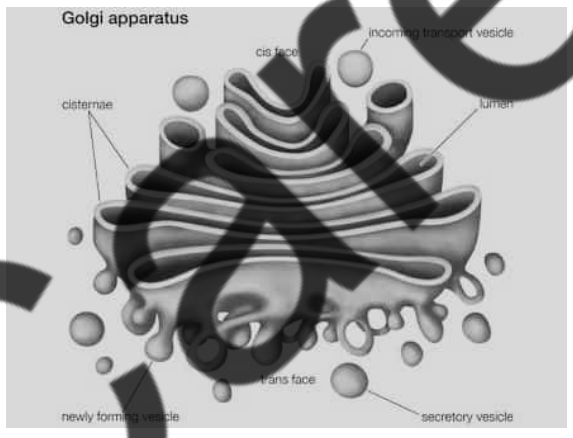


Figure depicts the structure of thylakoids and cisternae.

Q. 76 Cellular organelles with membranes are:

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Option 1:
chromosomes, ribosomes and endoplasmic reticulum

Option 2:
endoplasmic reticulum, ribosomes and nuclei

Option 3:
lysosomes, Golgi apparatus and mitochondria

Option 4:
nuclei, ribosomes and mitochondria

Correct Answer:
lysosomes, Golgi apparatus and mitochondria

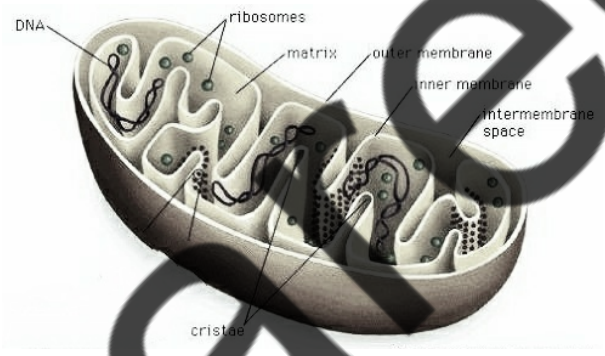
Solution:
As we learnt in

Mitochondria -

Each mitochondria is a double membrane bound structure with the outer membrane and the inner membrane dividing its lumen distinctly into two aqueous compartments.

- wherein

It is the site of aerobic respiration. It produces cellular energy in the form of ATP, hence called as 'power house' of the cell.



And,

Cell organelles in eukaryotic cell -

Mitochondria and golgi apparatus are usally present in animal cells. Microbodies including lysosomes or their equivalent are present.

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-

Cell organelles are membrane-bound structures.

Q. 77 Auxin can be bioassayed by:

Option 1:
Hydroponics

Option 2:
Potometer

Option 3:
Lettuce hypocotyl elongation

Option 4:
Avena coleoptile curvature

Correct Answer:
Avena coleoptile curvature

Solution:

Auxin -

Tryptophan is the precursor of Indole acetic acid (IAA, natural auxin). Important biassay of auxins are Avena curvature test and cross root inhibition test. discovered by Went.

Q. 78 Which of the following layers in an antral follicle is acellular ?

Option 1:
Theca interna

Option 2:
Stroma

Option 3:
Zona pellucida

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Option 4:
Granulosa

Correct Answer:
Zona pellucida

Solution:
As we learnt in

Zona pellucida -

The secondary oocyte forms a new membrane called zona pellucida surrounding it.

-

Zona pellucida, a glycoprotein membrane covering secondary oocyte, is acellular.

Q. 79 Satellite DNA is important because it:

Option 1:
shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

Option 2:
does not code for proteins and is same in all members of the population.

Option 3:
codes for enzymes needed for DNA replication.

Option 4:
codes for proteins needed in cell cycle.

Correct Answer:
shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

Solution:
As we learnt in

DNA fingerprinting -

The technique of DNA fingerprinting was initially developed by Alec Jeffery.

- wherein

He used a satellite DNA as probe that shows very high degree of polymorphism.

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Shows high degree of polymorphism in population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

Q. 80 Cell wall is absent in :

Option 1:
Funaria

Option 2:
Mycoplasma

Option 3:
Nostoc

Option 4:
Aspergillus

Correct Answer:
Mycoplasma

Solution:
As we learnt in

Prokaryotic Cells -

These are mainly represented by bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia Like Organisms).

-

Mycoplasm are prokaryotes and they lack cell wall.

Q. 81 In angiosperms, microsporogenesis and megasporogenesis:

Option 1:
form gametes without further divisions

Option 2:
Involve meiosis

Option 3:
occur in ovule

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Option 4:
occur in anther

Correct Answer:
Involve meiosis

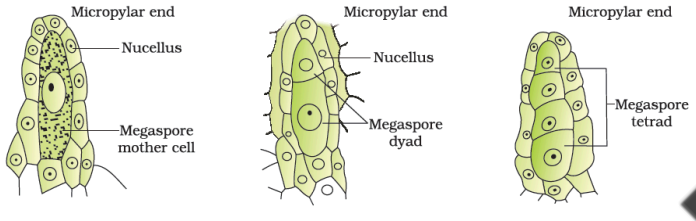
Solution:
As we learnt in

Megasporogenesis -

The process of formation of megaspores from the megaspore mother cell is called megasporogenesis. Ovules generally differentiate a single megaspore mother cell (MMC) in the micropylar region of the nucellus. It is a large cell containing dense cytoplasm and a prominent nucleus.

- wherein

Meiosis results in the production of four megaspores .



Microsporongenesis -

As the anther develops, the cells of the sporogenous tissue undergo meiotic divisions to form microspore tetrads. As each cell of the sprorogenous tissue is capable of giving rise to a microspore tetrad. Each one is a potential pollen or microspore mother cell.

- wherein

The process of formation of microspores from a pollen mother cell (PMC) through meiosis is called microsporogenesis.

Microsporogenesis & Megasporogenesis takes place through meiosis.

Hence, the correct option is (b).

Q. 82 Roots play insignificant role in absorption of water in:

Option 1:
Pistia

Option 2:
Pea

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Option 3:
Wheat

Option 4:
Sunflower

Correct Answer:
Pistia

Solution:
As we learnt in

Rate of Water Absorption (R) -

$$R\,\alpha\,\frac{1}{Salt\,concentration\,in\,soil}$$

$$R\,\alpha\,Soil\,temperature$$

$$R\,\alpha\,Soil\,aeration\,(O_2)$$

$$R\,\alpha\,Soil\,moisture$$

-

Rate of water absorption in the roots depends upon soil conditions. But Pistia is a hydrophytic plant, hence, roots has no significance in water absorption.

Q. 83 Which of the following are most suitable indicators of SO_2 pollution in the environment ?

Option 1:
Conifers

Option 2:
Algae

Option 3:
Fungi

Option 4:
Lichens

Correct Answer:
Lichens

Solution:
As we learnt in

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pioneer species -

The species that invade a bare area are called pioneer species.

- wherein

In primary succession on rocks these are usually lichens which are able to secrete acids to dissolve rock, helping in weathering and soil formation.

Lichens are widely used as environmental indicators or bio indicator. If air is badly polluted with SO_2 there may be no lichens present.

Q. 84 Grafted kidney may be rejected in a patient due to:

Option 1:
Cell-mediated immune response

Option 2:
Passive immune response

Option 3:
Innate immune response

Option 4:
Humoral immune response

Correct Answer:
Cell-mediated immune response

Solution:
As we learnt in

Transplantation -

Very often, when some human organs like heart, eye, liver, kidney fail to function satisfactorily, transplantation is the only remedy to enable the patient to live a normal life

- wherein

For this a suitable donor is required. Grafts from just any source – an animal, another primate, or any human beings cannot be made since the grafts would be rejected sooner or later. Tissue matching, blood group matching are essential before undertaking any graft/transplant and even after this the patient has to take immuno–suppresants all his/her life. The body is able to differentiate 'self ' and 'nonself' and the cell-mediated immune response is responsible for the graft rejection

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Grafted kidney may be rejected in a patient due to the cell-mediated immune response that is mediated by T-lymphocytes.

Q. 85 Body having meshwork of cells, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum:

Option 1:
Porifera

Option 2:
Mollusca

Option 3:
Protozoa

Option 4:
Coelenterata

Correct Answer:
Porifera

Solution:
As we learnt in

Phylum porifera -

Most primitive multicellular organism having pores in their body walls.(Known as ostia).They are mostly marine, few freshwater.The only phylum without nerves cells.

- wherein

Spongilla,Sycon,Euspongia.

In Porifera (sponges), bodies are asymmetrical. Body lacks tissues or organs, but from a meshwork of cells surrounding channels that open to the outside through pores, and that expand into internal cavities lined with food filtering flagellated cells (choanocytes).

Q. 86 In which group of organisms the cell walls form two thin overlapping shells which fit together ?

Option 1:
Euglenoids

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Option 2:
Dinoflagellates

Option 3:
Slime moulds

Option 4:
Chrysophytes

Correct Answer:
Chrysophytes

Solution:
As we learnt in

Diatomas / Chrysophytes -

1. May be present in marine, fresh water or terrestrial habitats.
2. Movement is brought by mucilage propulsion.

- wherein

1. Serve as sewage pollution indicator.
2. Cell walls are embedded with silica, which are indestructible and thus, forms diatomaceous earth or kieselguhr.

Chrysophytes are photosynthetic protists. They have overlapping cell wall like soap box

---

Q. 87 Choose the **wrong** statement:

Option 1:
Neurospora is used in the study of biochemical genetics

Option 2:
Morels and truffles are poisonous mushrooms

Option 3:
Yeast is unicellular and useful in fermentation

Option 4:
Penicillium is multicellular and produces antibiotics

Correct Answer:
Morels and truffles are poisonous mushrooms

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Solution:
As we learnt in

Single Cell Protein (SCP) -

Single-cell protein (SCP) refers to edible unicellular microorganisms.

- wherein

The biomass or protein extract from pure or mixed cultures of algae, yeasts, fungi or bacteria may be used as an ingredient or a substitute for protein-rich foods, and is suitable for human consumption or as animal feeds.

Truffles and morels are the both edible fungi, or mushrooms, but they are not the same thing. Truffles are the body of a mushroom that grows underground.

Q. 88 In human females, meiosis-II is not completed until ?

Option 1:
fertilization

Option 2:
uterine implantation

Option 3:
birth

Option 4:
puberty

Correct Answer:
fertilization

Solution:
Each oocyte stays in meiosis II metaphase after ovulation. Only if fertilisation takes place, producing a mature, fertilised ovum and the second polar body, is meiosis II complete.

Q. 89 Eutrophication of water bodies leading to killing of fishes is mainly due to non-availability of:

Option 1:
light

Option 2:
essential minerals

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Option 3:
oxygen

Option 4:
food

Correct Answer:
oxygen

Solution:

- **Eutrophication** refers to the excessive growth of algae, plants, and animals in water bodies as a result of the enrichment of nutrients, especially nitrogen and phosphate.
- In water, **oxygen replenishment decreases dramatically.**
- **Non-availability of oxygen** leads to the death of aquatic organisms.

Hence, the correct answer is **option 3).**

Q. 90 The enzyme that is not present in succus entericus is:

Option 1:
nucleases

Option 2:
nucleosidase

Option 3:
lipase

Option 4:
maltase

Correct Answer:
nucleases

Solution:
Nucleases capable of digesting nucleic acid molecules are not found in intestinal fluid. These include endonucleases and exonucleases that are not found in the small intestine. Hence, option 1 is correct.

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